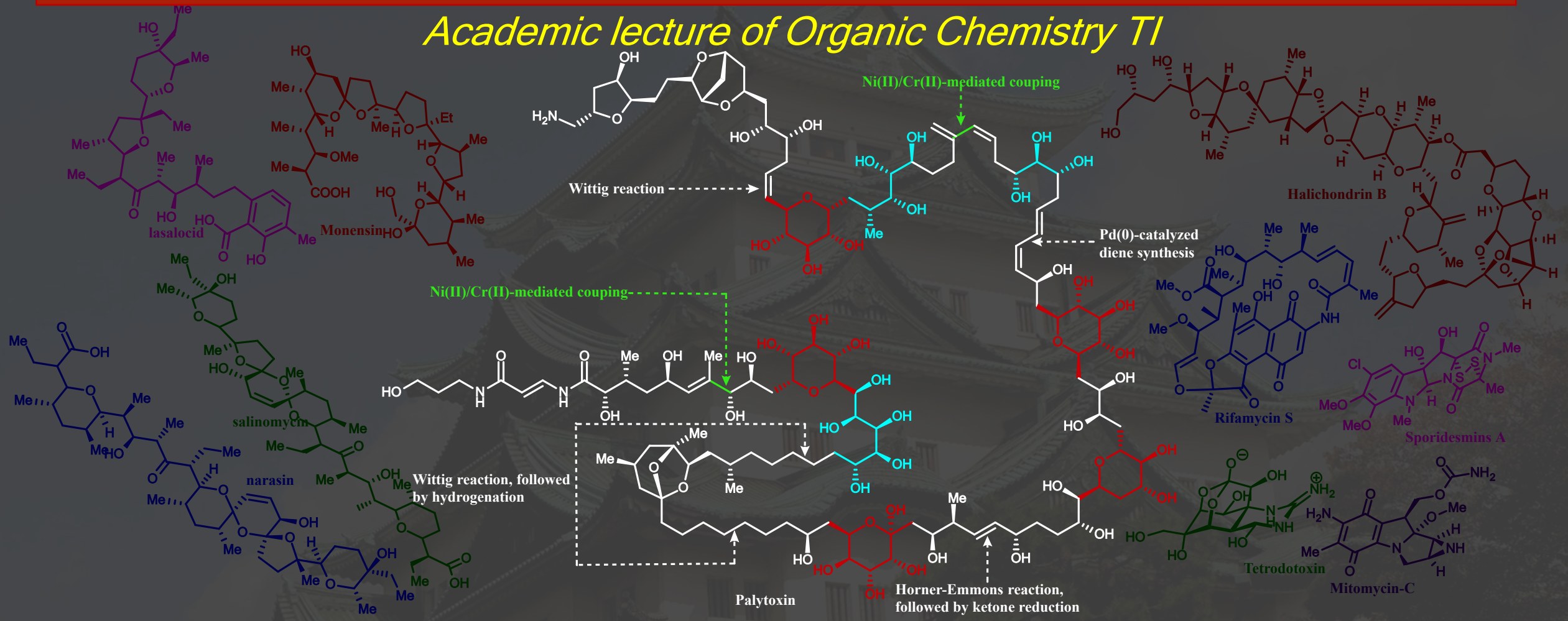
Review of Yoshito Kishi's academic achievements

Academic lecture of Organic Chemistry TI



"It is not unreasonable to say that Kishi has become a legend in his own time."—Robert M. Williams

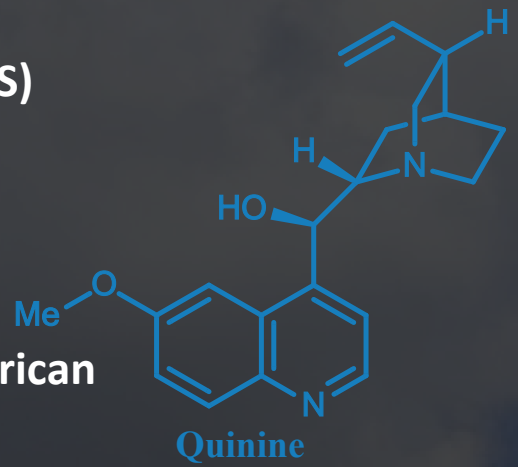
Zhang Shuo

College of Chemistry, Jilin University

Robert Burns Woodward

Selected Awards

- 1945: John Scott Medal, from the Franklin Institute and City of Philadelphia
- 1955: Leo Hendrik Baekeland Award, from the North Jersey Section of the American Chemical Society
- 1956: Elected a Foreign Member of the Royal Society (ForMemRS)
- 1959: Davy Medal, from the Royal Society
- 1961: Roger Adams Medal, from the American Chemical Society
- 1964: National Medal of Science from the United States
- 1965: Nobel Prize in Chemistry
- 1967: Willard Gibbs Award from the Chicago Section of the American Chemical Society
- 1968: Lavoisier Medal from the Société chimique de France
- 1969: Pius XI Gold Medal, from the Pontifical Academy of Sciences
- 1970: The Order of the Rising Sun, Second Class from the Emperor of Japan
- 1970: Hanbury Memorial Medal from The Pharmaceutical Society of Great Britain
- 1970: Pierre Bruylants Medal from the University of Louvain
- 1971: AMA Scientific Achievement Award
- 1973: Cope Award from the American Chemical Society, shared with Roald Hoffmann
- 1978: Copley Medal from the Royal Society, London



Quinine



Born :1917 •
B.S. Massachusetts Institute of Technology :1936 •
Ph.D. Massachusetts Institute of Technology :1937 •
Assistant Professor, Harvard University :1944 •
Associate Professor, Harvard University :1946 •
Full Professor, Harvard University :1950 •
Died :1979 •

https://en.wikipedia.org/wiki/Robert_Burns_Woodward

Selected synthesis by the Woodward Group



Quinine

Reserpine

Chlorophyll A

Cephalosporin C

Vitamin B12

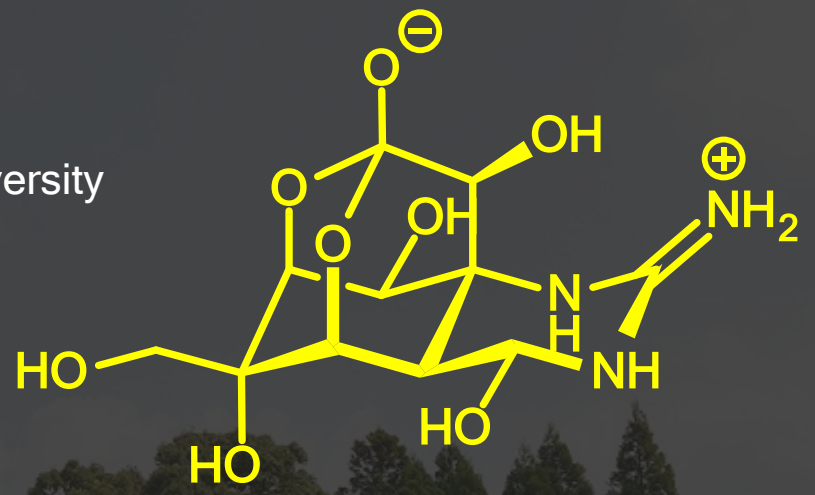
Strychnine

Erythromycin A

Hirata Yoshimasa



- 1915: Born in Yamaguchi city
- 1941: B.S. Chem, Tokyo Imperial University
- 1943: Assistant Professor, Tokyo Imperial University
- 1944: Associate Professor, Nagoya University
- 1952: Visiting scholar, Harvard University
- 1954: Full Professor, Nagoya University
- 2000: Died



Tetrodotoxin

Selected Awards

- 1941: Chunichi Cultural Award
- 1977: Japan Academy Prize
- 1977: Fujihara Award
- 1987: Orders of the Sacred Treasure
- 1996: Nakanishi Prize

Educator, organic chemist, natural product chemist
Emeritus Professor of Nagoya University
Major research efforts: Separation and identification of natural products

https://zh.wikipedia.org/wiki/ 平田義

Tohru Fukuyama

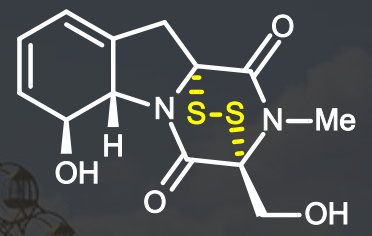


- 1948: Born in Ancho, Aichi
- 1971: B.S. Chem, Nagoya University
- 1973: M.S. Chem, Nagoya University
- 1977: Ph.D. Yoshito Kishi, Harvard University
- 1977 - 1978: Postdoctoral Associate with Y. Kishi
- 1978: Assistant Professor, Rice University
- 1982: Associate Professor, Rice University
- 1988: Full Professor, Rice University
- 1995: Professor of Pharmaceutical Sciences, University of Tokyo
- 2012: Professor at Nagoya University

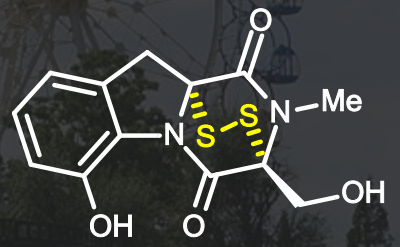
Selected Awards

- 1993: ACS Arthur C. Cope Scholar Award
- 2002: Synthetic Organic Chemistry Award, Japan
- 2003: ISHC Senior Award in Heterocyclic Chemistry
- 2004: ACS Award for Creative Work in Synthetic Organic Chemistry
- 2006: The PSJ Award
- 2009: Medals of Honor

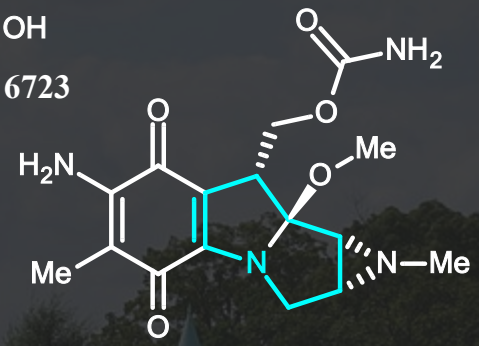
The Kishi years



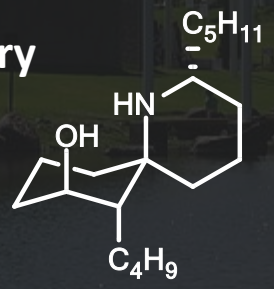
Gliotoxin JACS 1976 6723



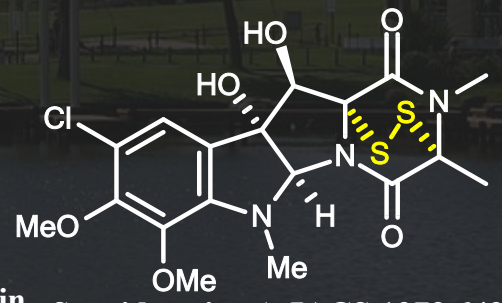
Dehydrogliotoxin JACS 1973 6492



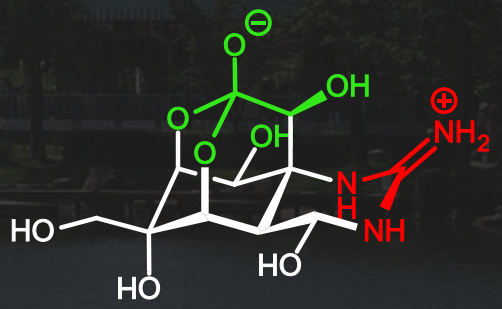
Porflcomycin JACS 1977 8155



Perhydrohistrionicotoxin JOC 1975 2009



Sporidesmins A JACS 1973 6493



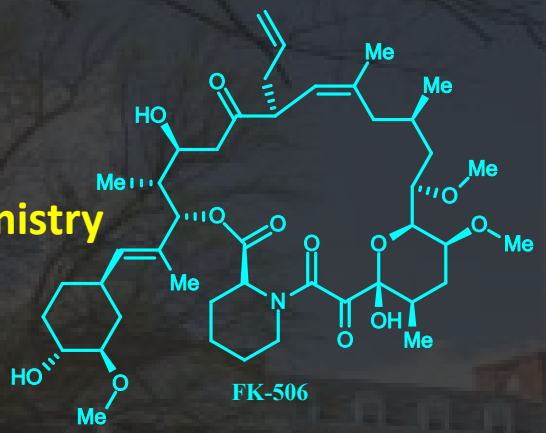
Tetrodotoxin JACS 1972 9219

https://en.wikipedia.org/wiki/Tohru_Fukuyama

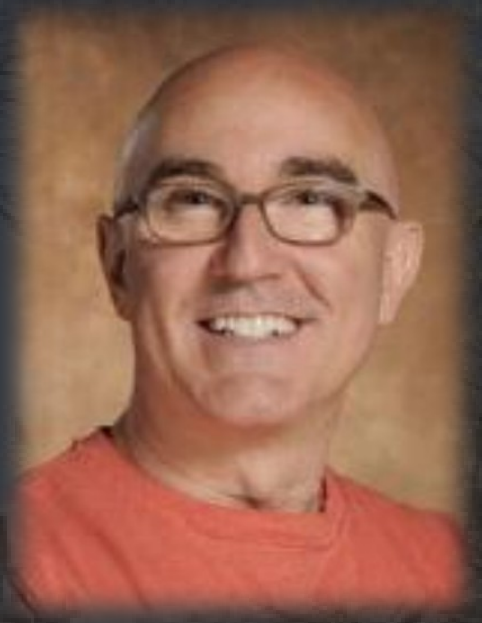
Stuart L. Schreiber

Selected Awards

- 1989: ACS Award in Pure Chemistry
- 1992: Ciba-Geigy Drew Award for Biomedical Research : Molecular Basis for Immune Regulation
- 1993: Leo Hendrik Baekeland Award, North Jersey Section of ACS
- 1993: Eli Lilly Award in Biological Chemistry, ACS
- 1994: American Chemical Society Award in Synthetic Organic Chemistry
- 1994: George Ledlie Prize (Harvard University)
- 1994: Paul Karrer Gold Medal
- 1995: Harrison Howe Award
- 1995: Warren Triennial Award (shared with Leland Hartwell)
- 1997: Tetrahedron Prize for Creativity in Organic Chemistry
- 2000: ACS Award for Bioorganic Chemistry
- 2001: William H. Nichols Medal
- 2001: Chiron Corporation Biotechnology Research Award, American Academy of Microbiology
- 2004: Society for Biomolecular Screening Achievement Award
- 2004: American Association of Cancer Institutes
- 2014: Arthur C. Cope Award
- 2015: Nagoya Gold Medal
- 2016: Wolf Prize



FK-506

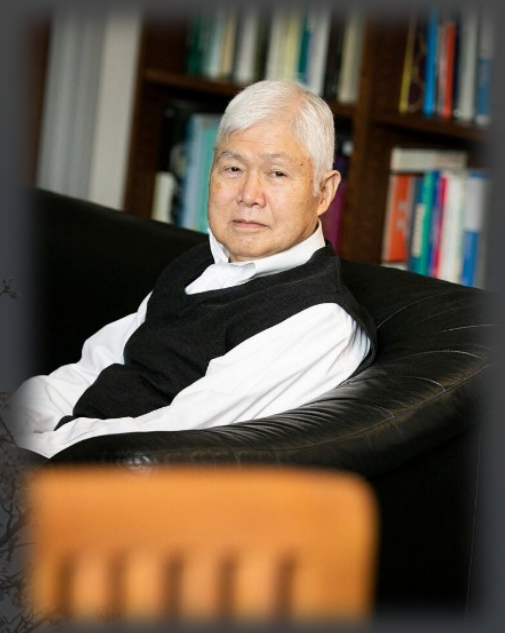


Born :1956 •
B.S. University of Virginia :1977 •
Ph.D. R.B.Woodward, Yoshito Kishi, Harvard University :1981 •
Assistant Professor, Yale University :1981 •
Associate Professor, Yale University :1984 •
Full Professor, Yale University :1986 •
Morris Loeb Professor, Harvard University :1988 •

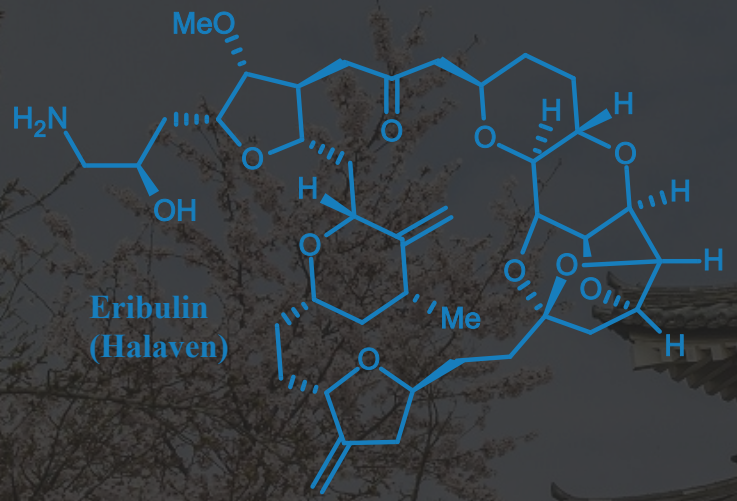
"He is the co-Founder of the Broad Institute. He has been active in chemical biology, especially the use of small molecules as probes of biology and medicine ."

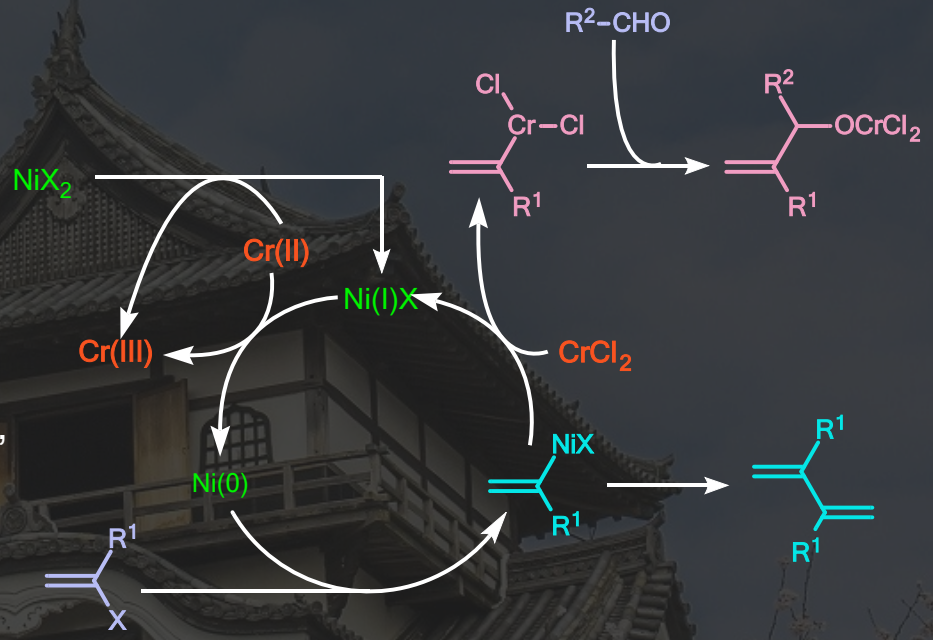
https://en.wikipedia.org/wiki/Stuart_Schreiber

Yoshito Kishi



- 1937: Born
- 1961: B.S. Chem, Nagoya University
- 1966: M.S. Chem, Nagoya University
- 1968: Postdoctoral Associate with R.B. Woodward, Harvard University
- 1974: Full Professor, Harvard University
- 1982: Morris Loeb Professor of Chemistry, Harvard University
- 1989: Chairman for the Department of Chemistry, Harvard University
- 1997: Morris Loeb Research Professor of Chemistry, Harvard University





Selected Awards

- 1980: ACS Award for Creative Work in Synthetic Organic Chemistry
- 1981: Harrison Howe Award
- 1988: Javits Neuroscience Investigator Award
- 1999: Imperial Prize of the Japan Academy
- 2001: Tetrahedron Prize
- 2001: Ernest Guenther Award
- 2001: the ACS Ernest Guenther Award in the Chemistry of Natural Products
- 2012: Yamada-Koga prize
- 2018: Ryoji Noyori Prize

https://en.wikipedia.org/wiki/Yoshito_Kishi

Selected synthesis by the Kishi Group



salinomycin

narasin

Halichondrin B

Rifamycin S

Sporidesmins A

Dehydrogliotoxin

lasalocid

Wittig reaction

Ni(II)/Cr(II)-mediated couping

Pd(0)-catalyzed diene synthesis

Ni(II)/Cr(II)-mediated couping

Mitomycin-C

Gliotoxin

Monensin

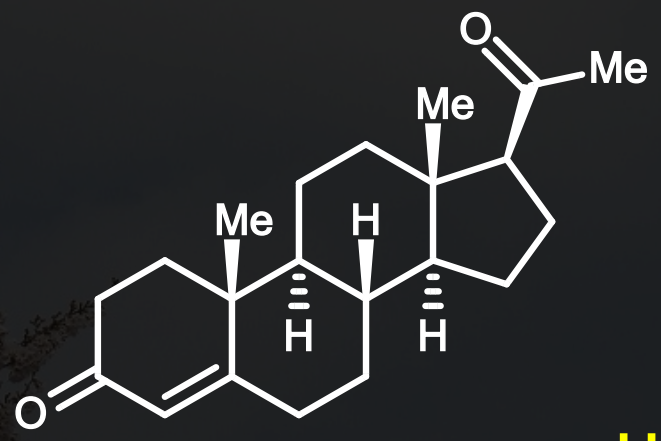
Wittig reaction, followed by hydrogenation

Palytoxin

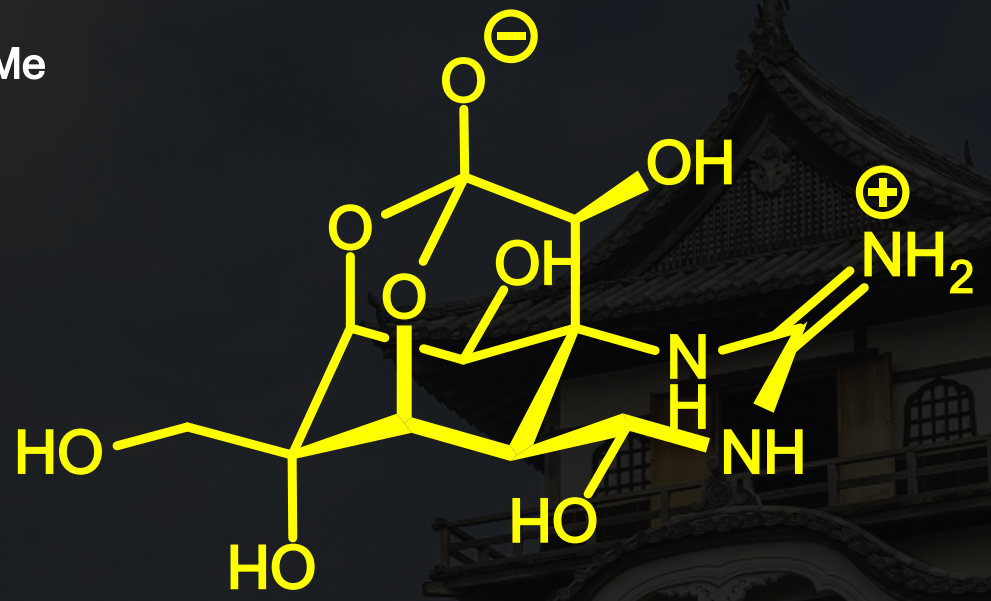
Horner-Emmons reaction, followed by ketone reduction

Tetrodotoxin

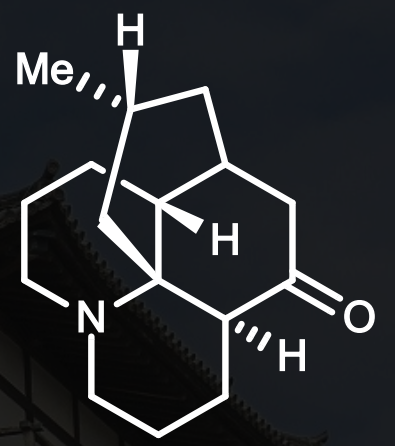
Difference between tetrodotoxin and others



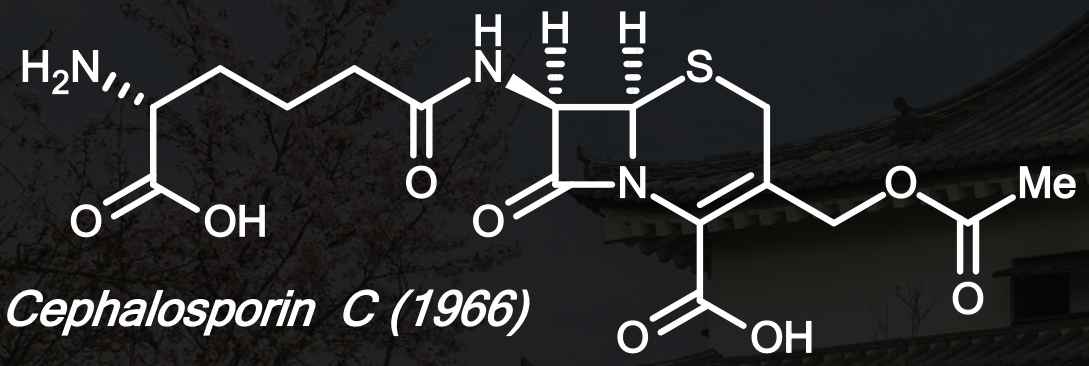
Progesterone (1971)



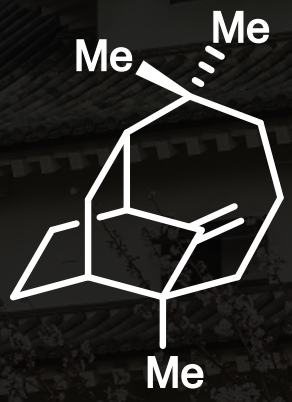
Tetrodotoxin



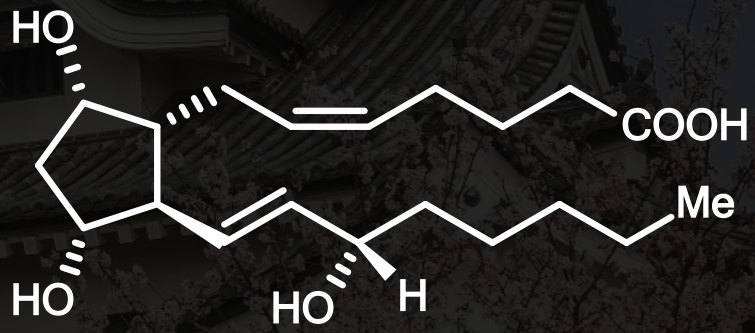
Lycopodine (1968)



Cephalosporin C (1966)

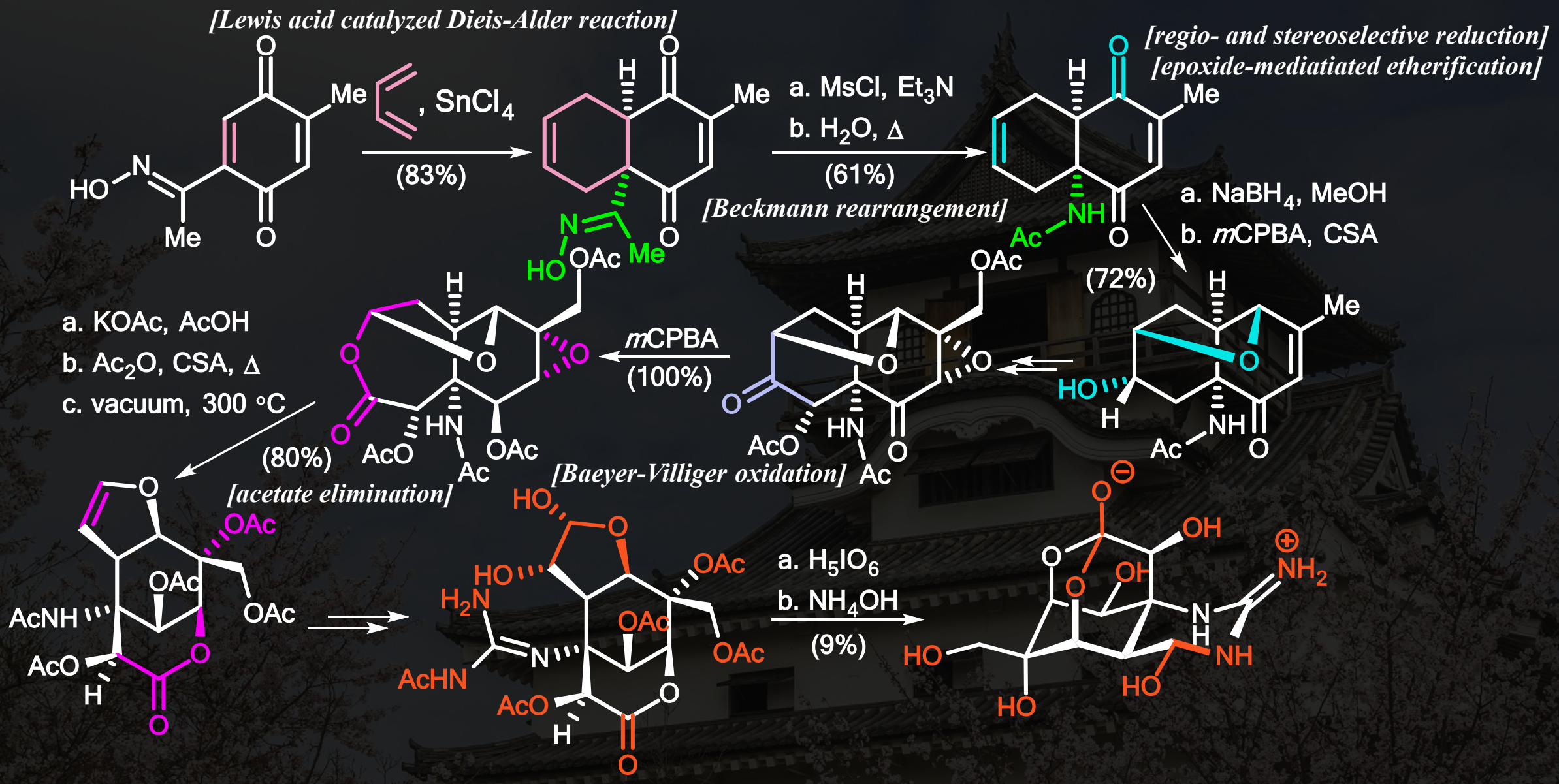


Longifolene (1961)



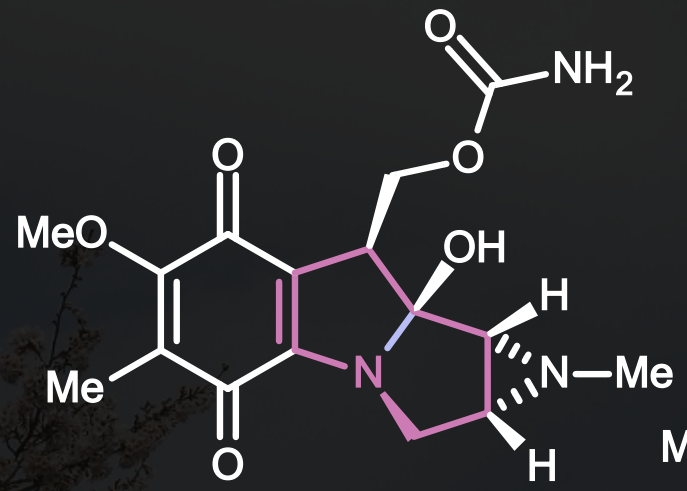
(±)-PGF_{2a} (1969)

Total synthesis of (±) – Tetrodotoxin

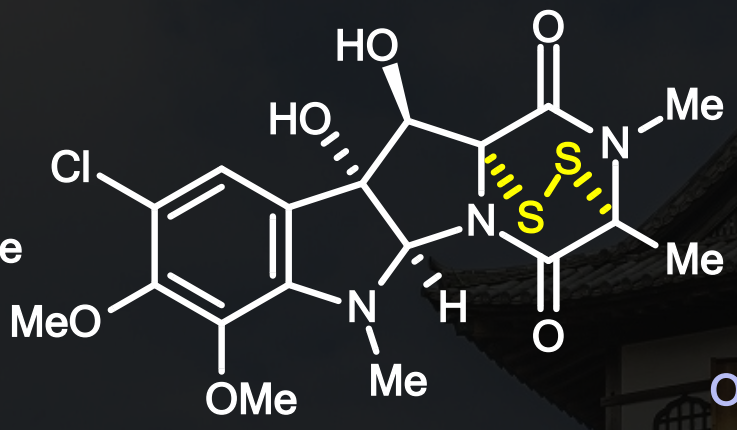


Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

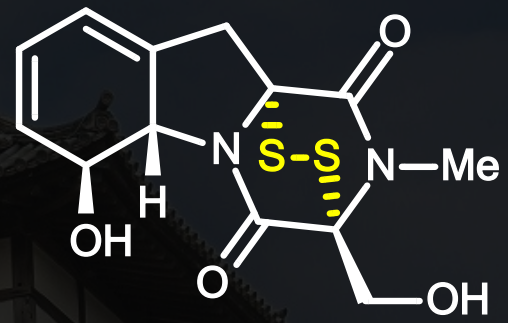
The brilliant daring-do of Kishi



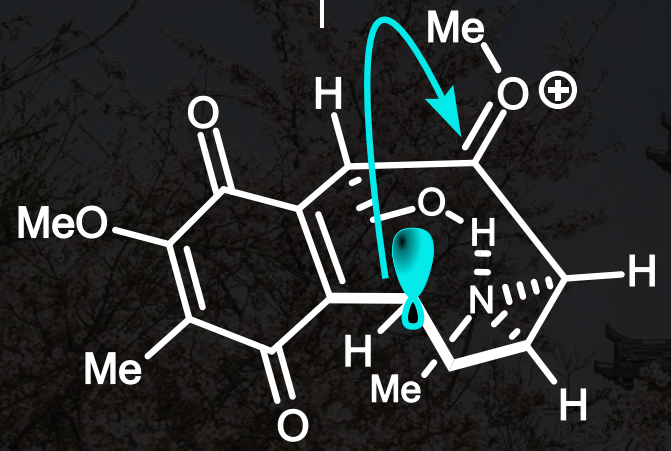
Mitomycn A

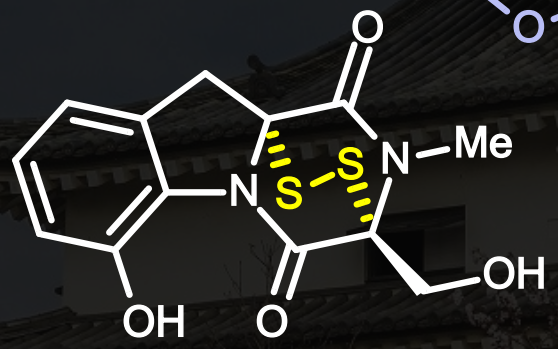


Sporidesmins A

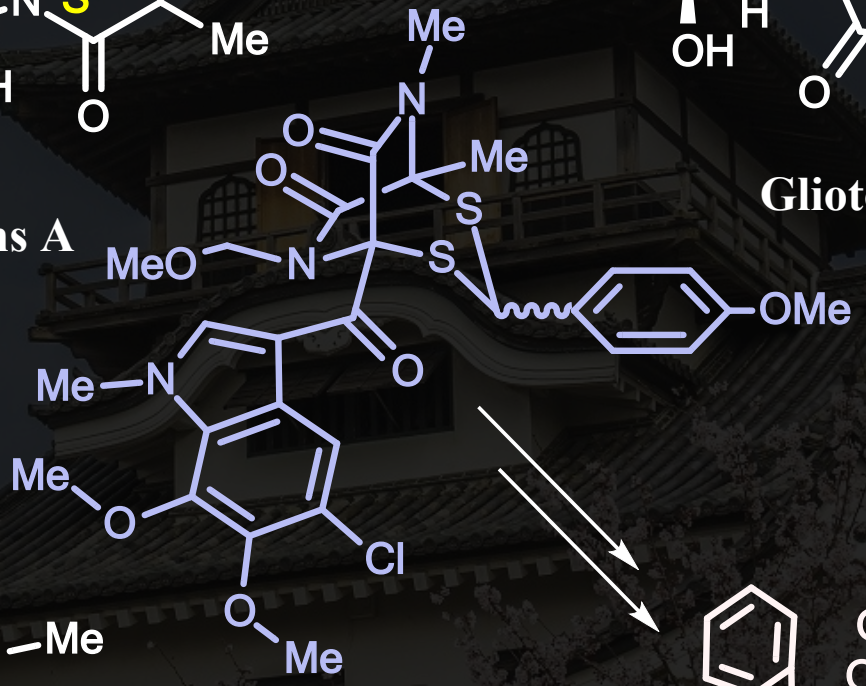


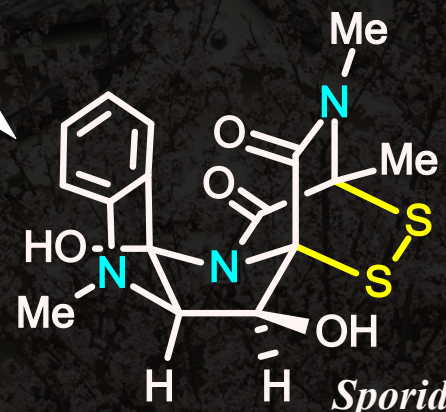
Gliotoxin





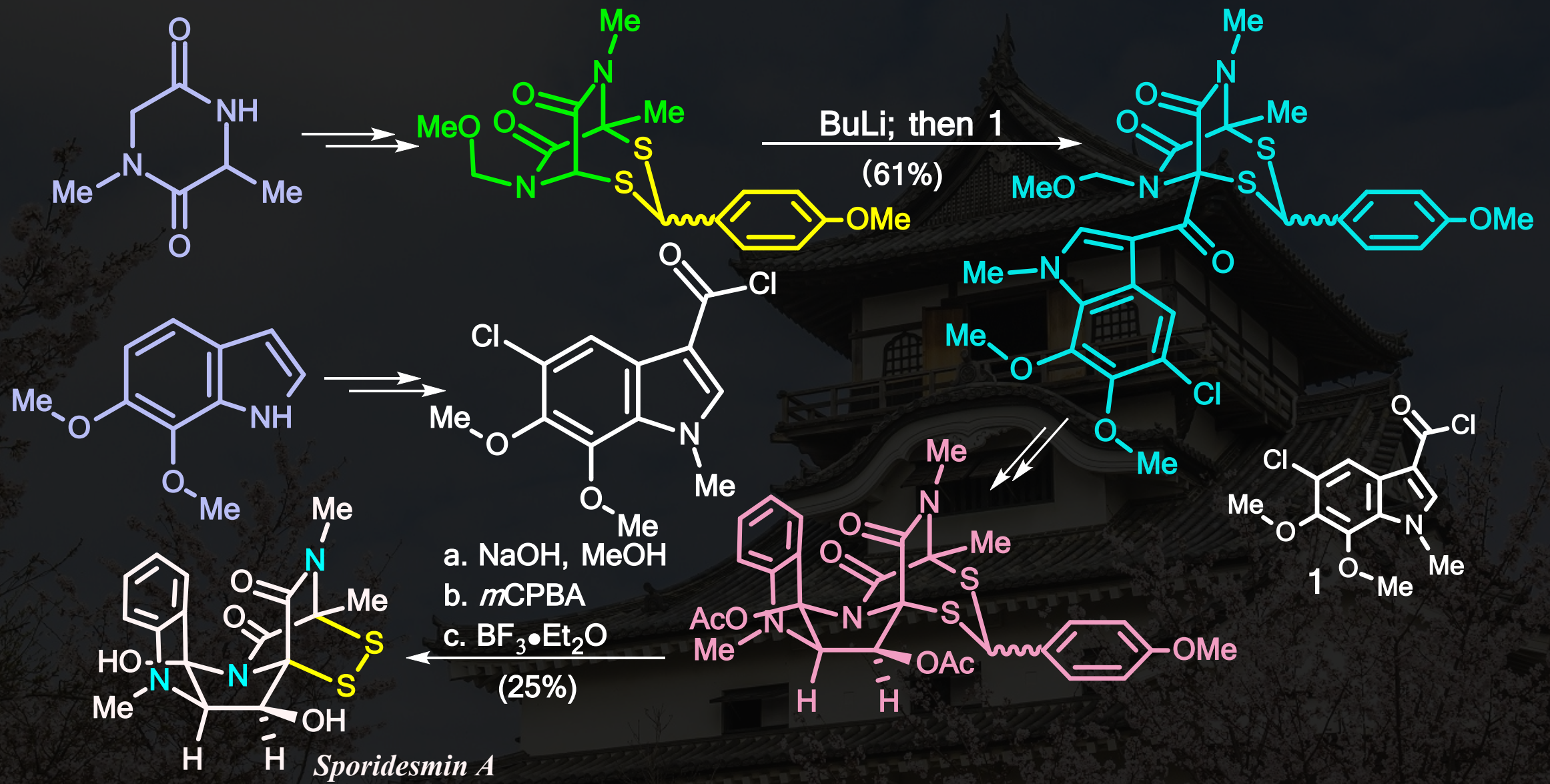
Dehydrogliotoxin





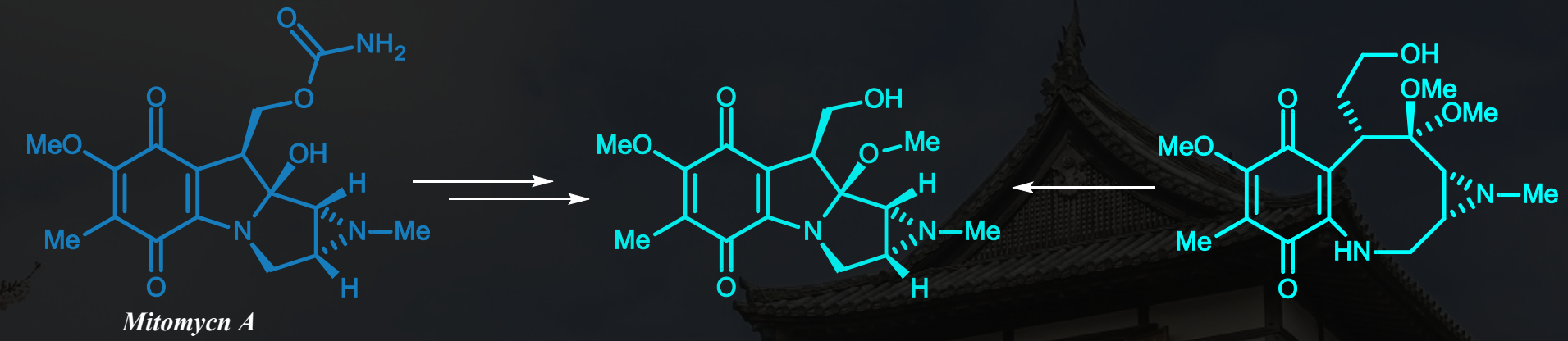
Sporidesmin A

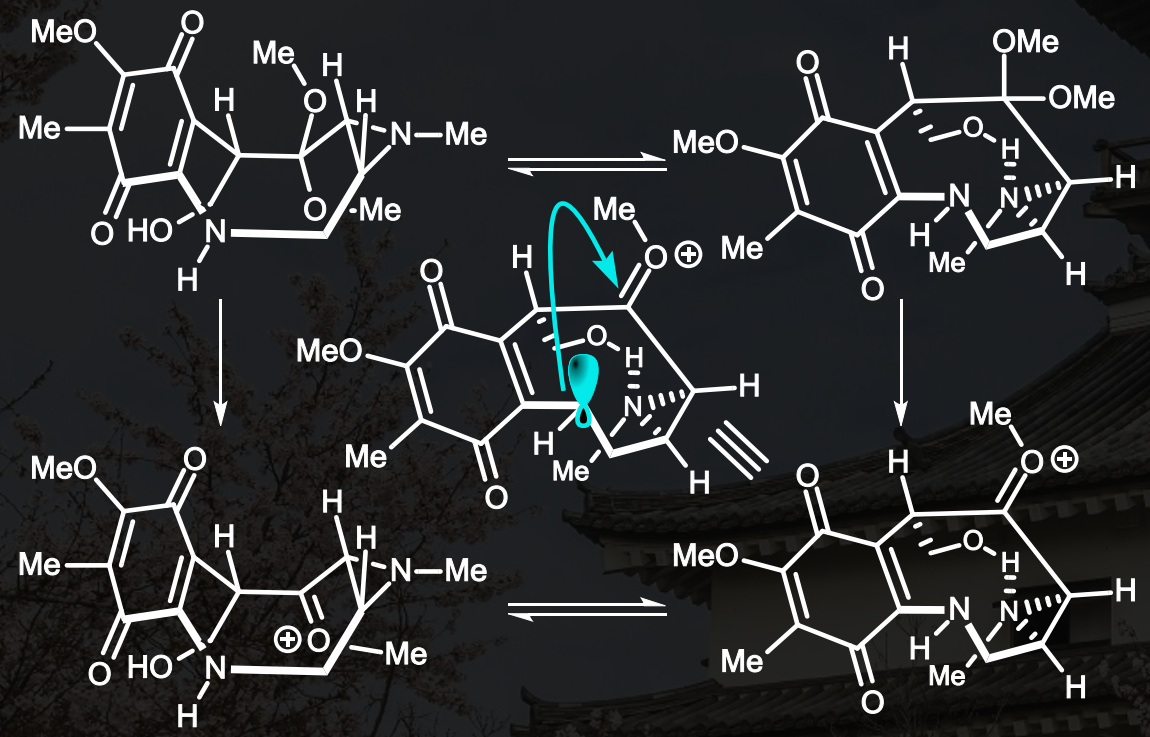
Total synthesis of Sporidesmin A



Kishi, Y.; Nakatsuka, S.; Fukuyama, T.; Havel, M. *Journal of the American Chemical Society* **1973**, *95*, 6493.

Total synthesis of Mitomycn A

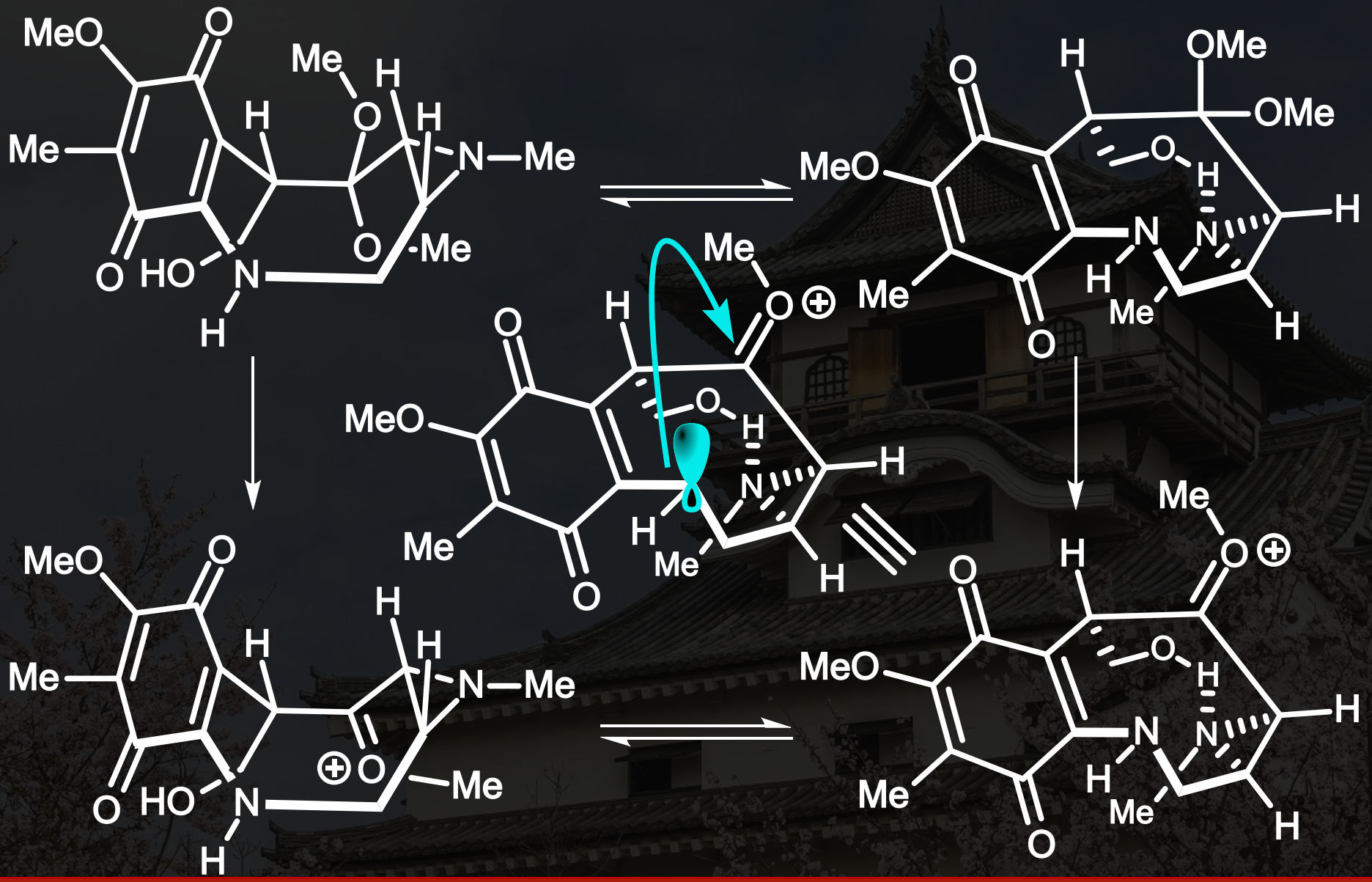




"These unstable alkaloids were constructed by synthesizing an eight-membered ring precursor, (a daring and important achievement at the time in it's own right) and performing a very delicate transannular cyclization to close the labile bicyclo[3.3.0] carbinolamine functional group. "

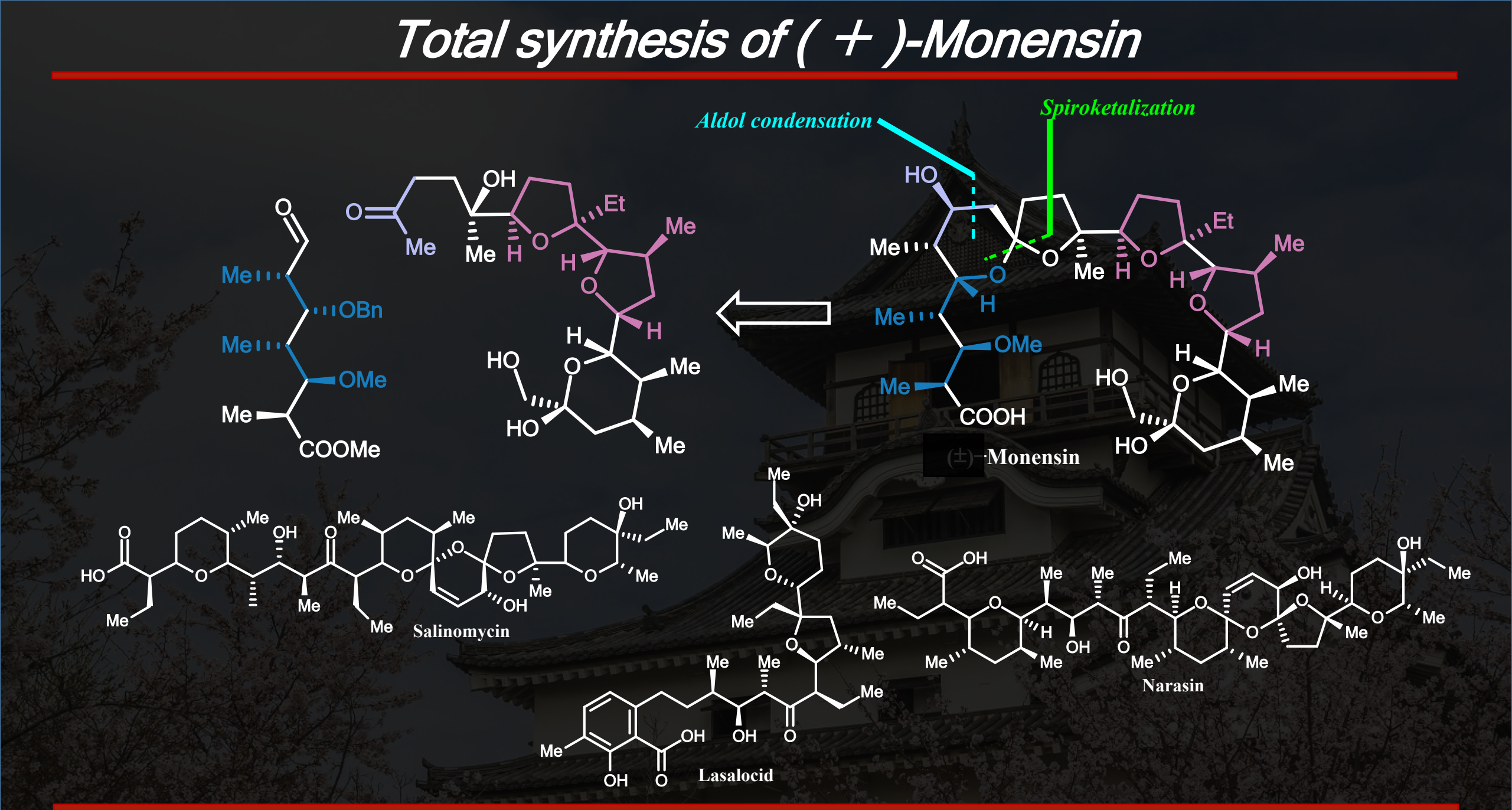
Kishi, Y. *Journal of Natural Products* **1979**, *42*, 549.

Total synthesis of Mitomycn A



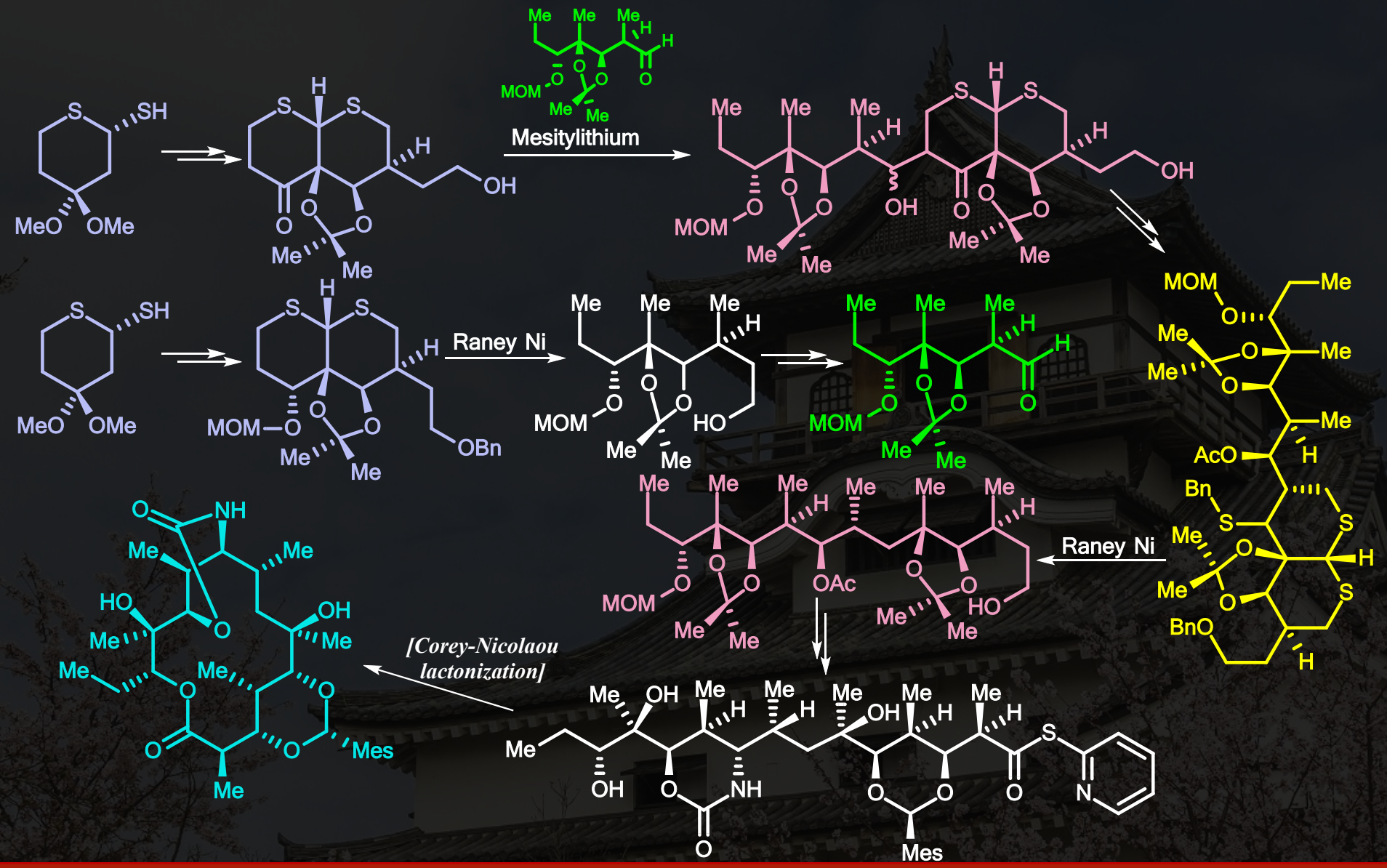
Kishi, Y. *Journal of Natural Products* **1979**, *42*, 549.

Total synthesis of (+)-Monensin



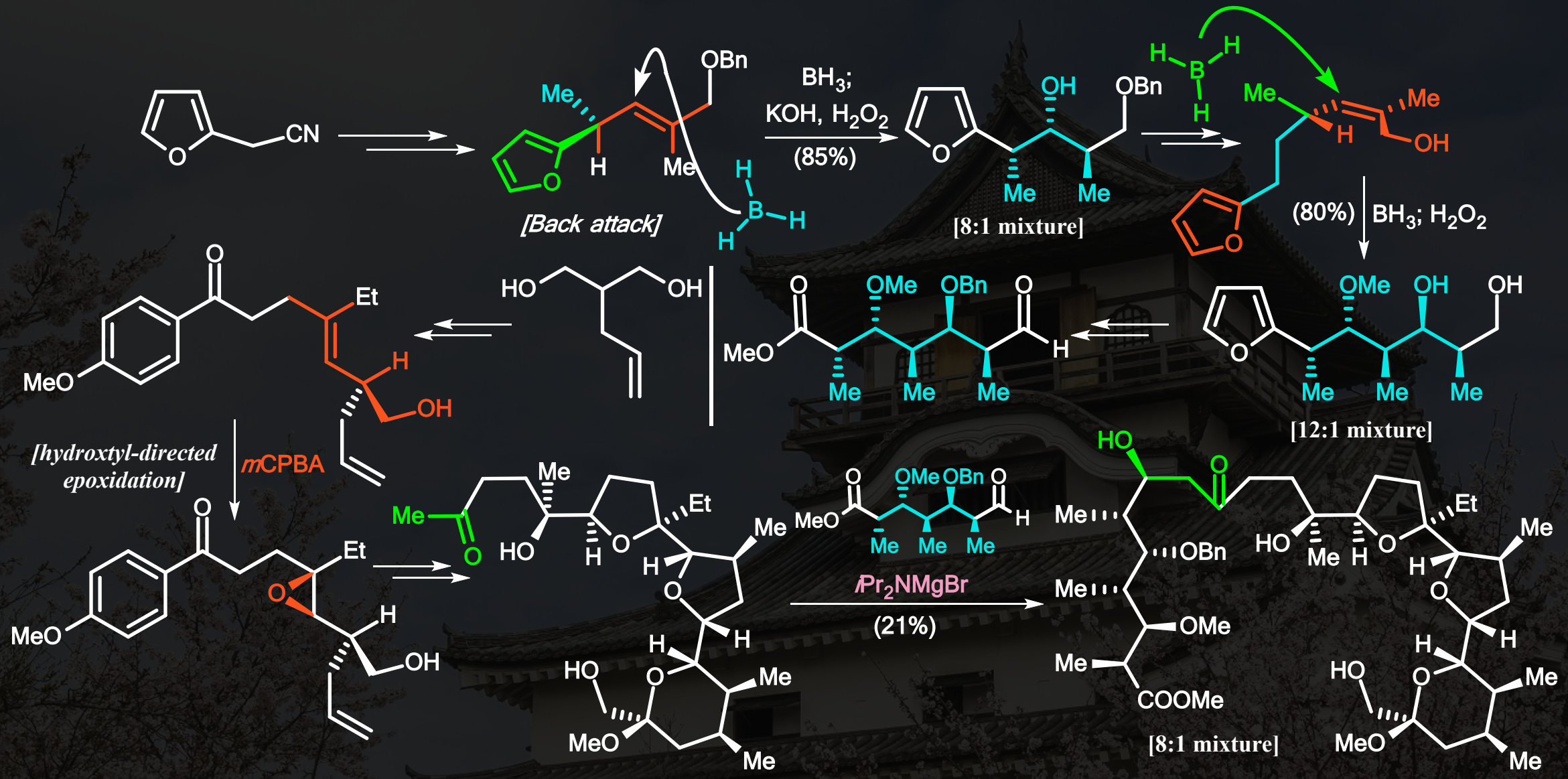
Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

Total synthesis of Erythromycin



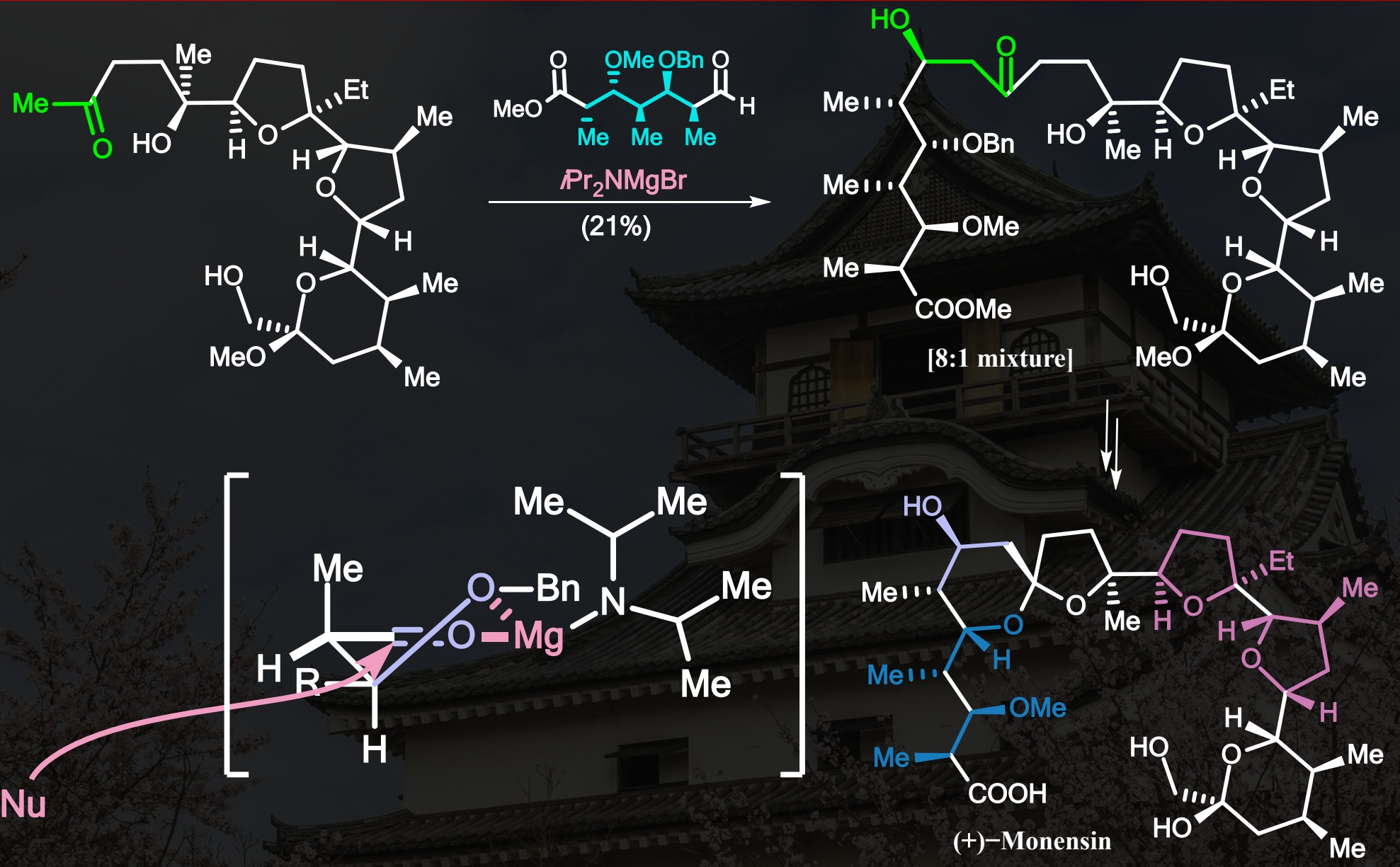
Woodward, R.; Logusch, E.; Nambiar, K.; Sakan, K.; Ward, D.; Au-Yeung, B.; Balaram, P.; Browne, L.; Card, P.; Chen, C. *Journal of the American Chemical Society* **1981**, *103*, 3215.

Total synthesis of (+)-Monensin



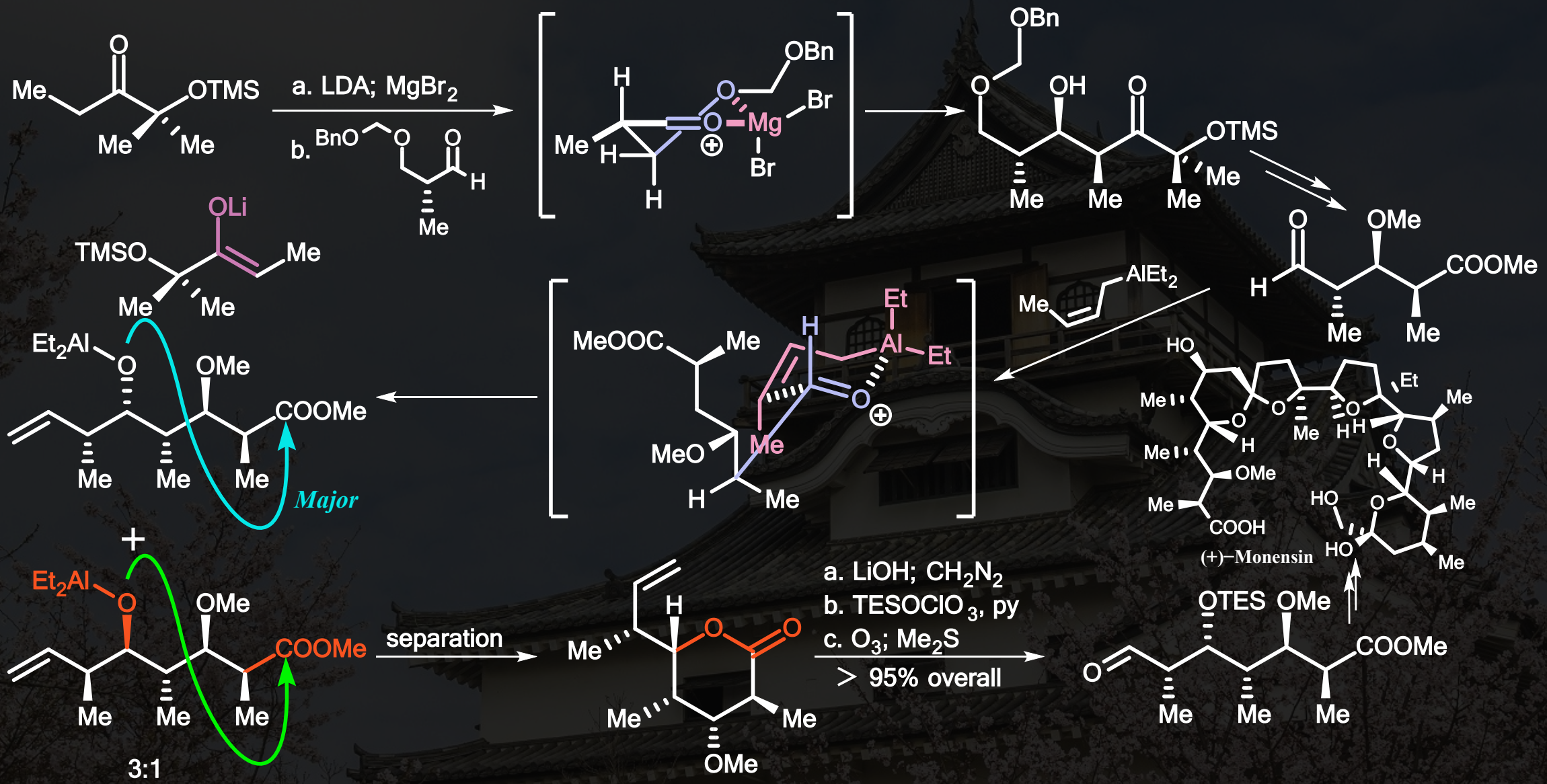
Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

Total synthesis of (+)-Monensin



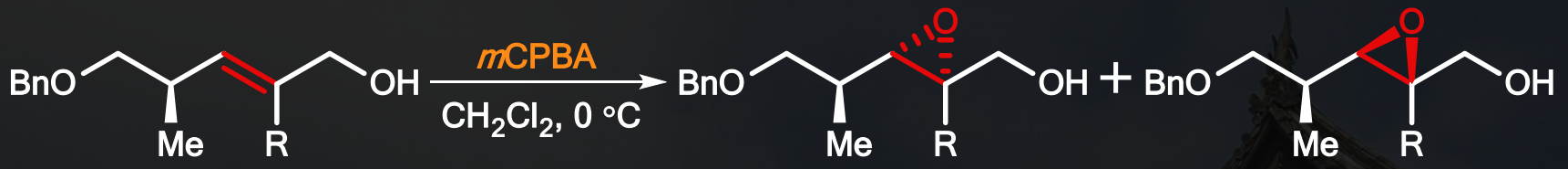
Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

Total synthesis of (+)-Monensin

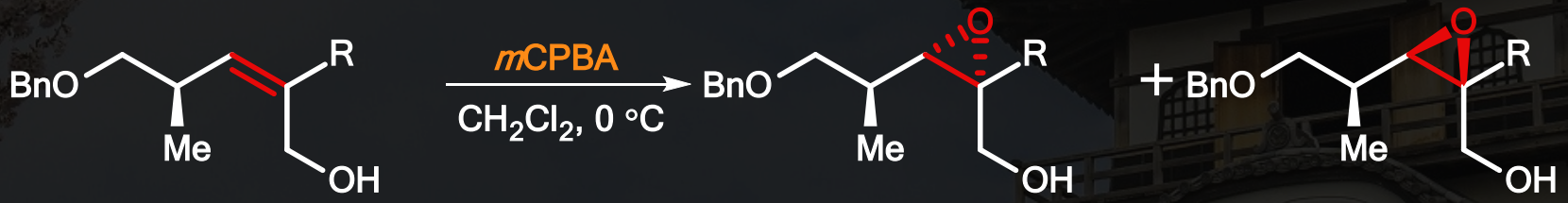


Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

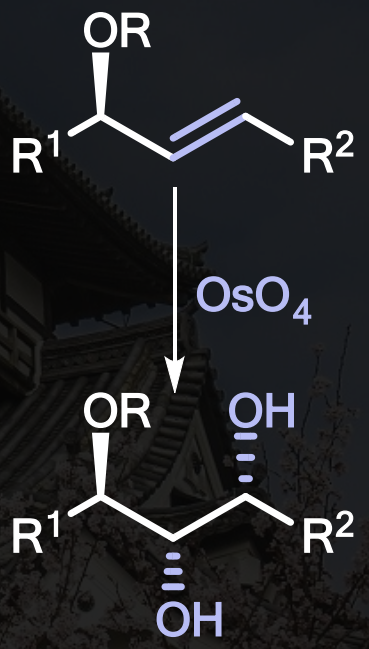
Chemistry from Monensin

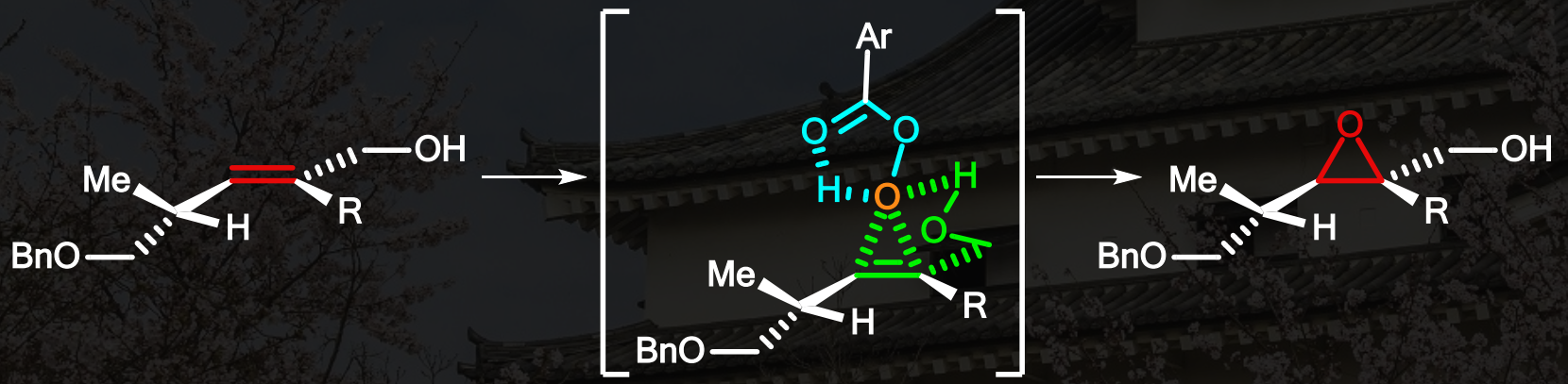


Stereoselectivity				
	R = Me :	> 20	:	1
	R = H :	3	:	2
	R = TMS :	> 20	:	1



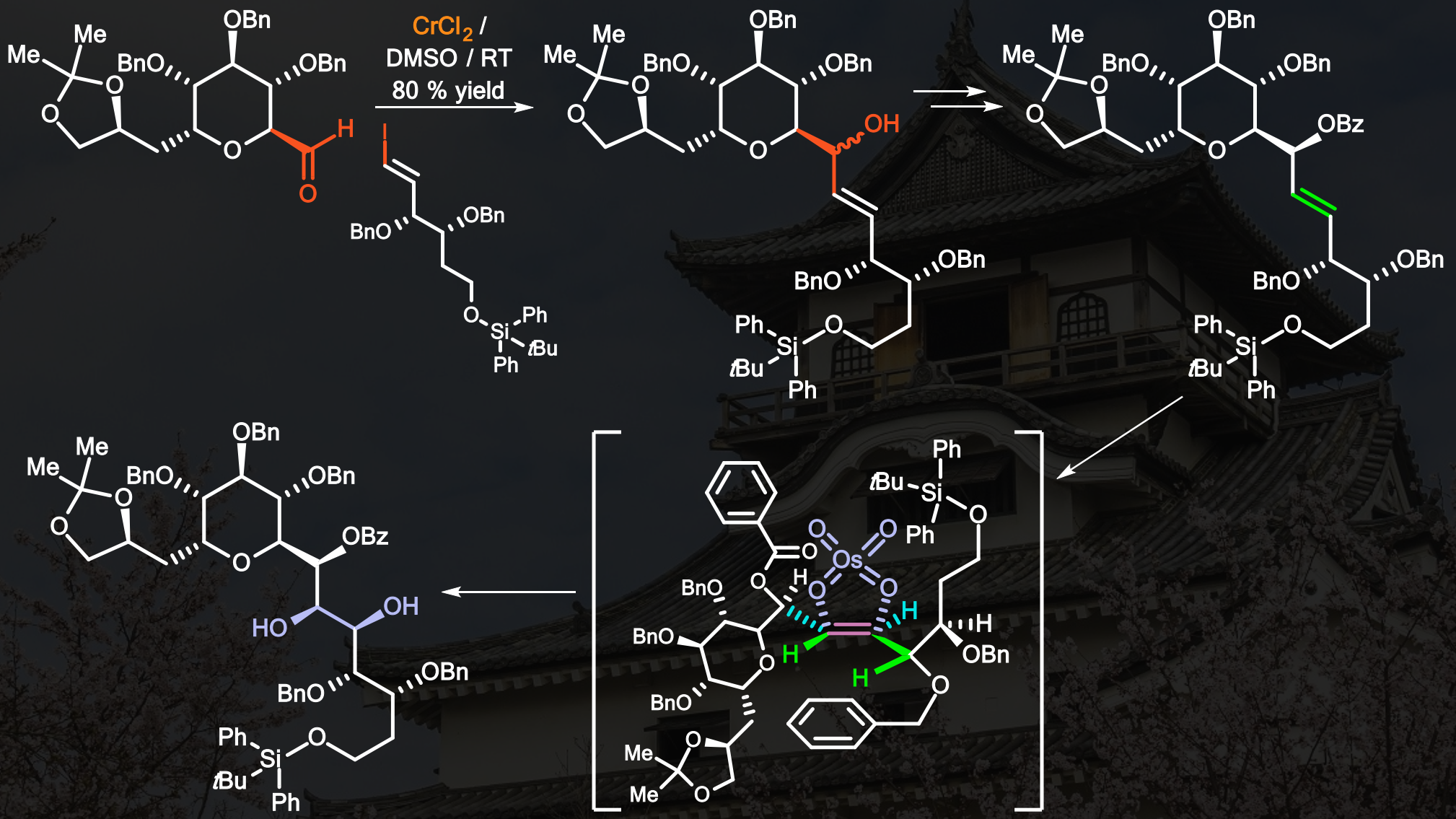
Stereoselectivity				
	R = Me :	> 20	:	1
	R = H :	> 20	:	1





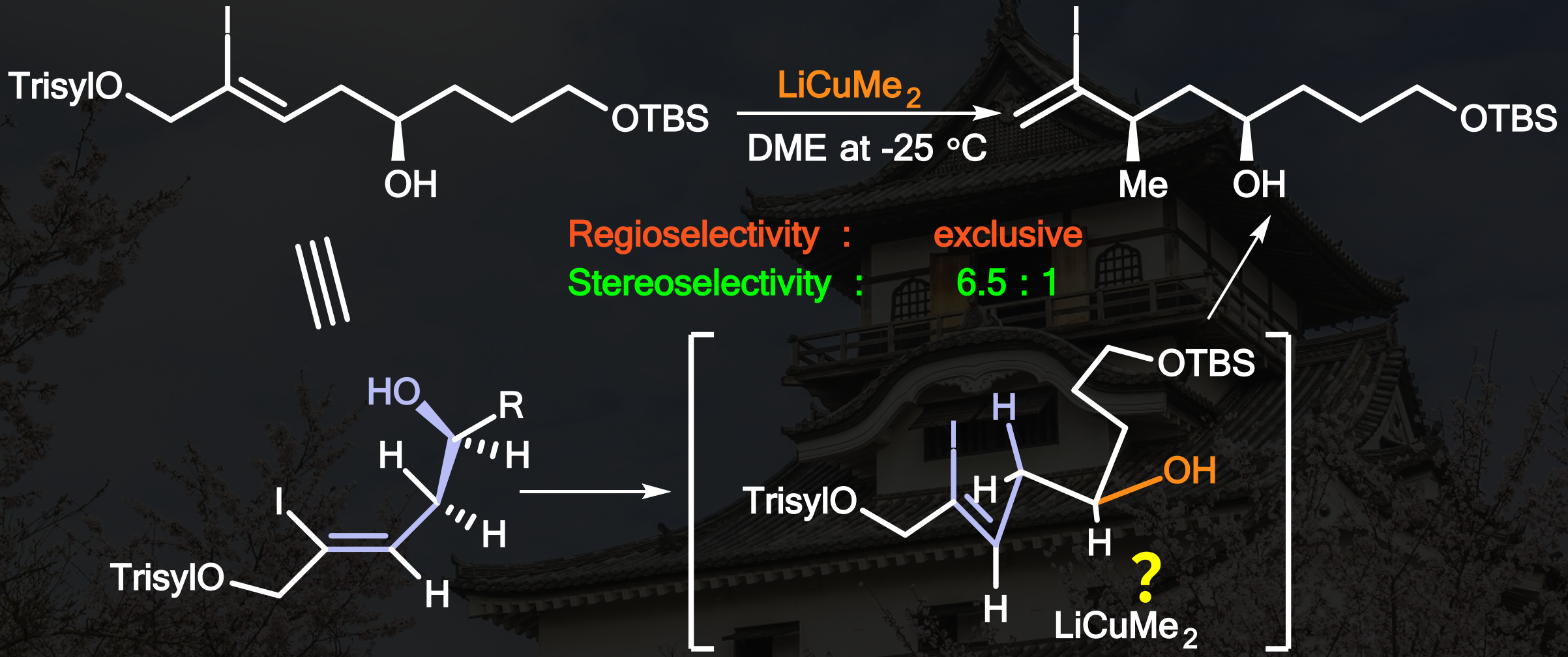
Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

Chemistry from Monensin



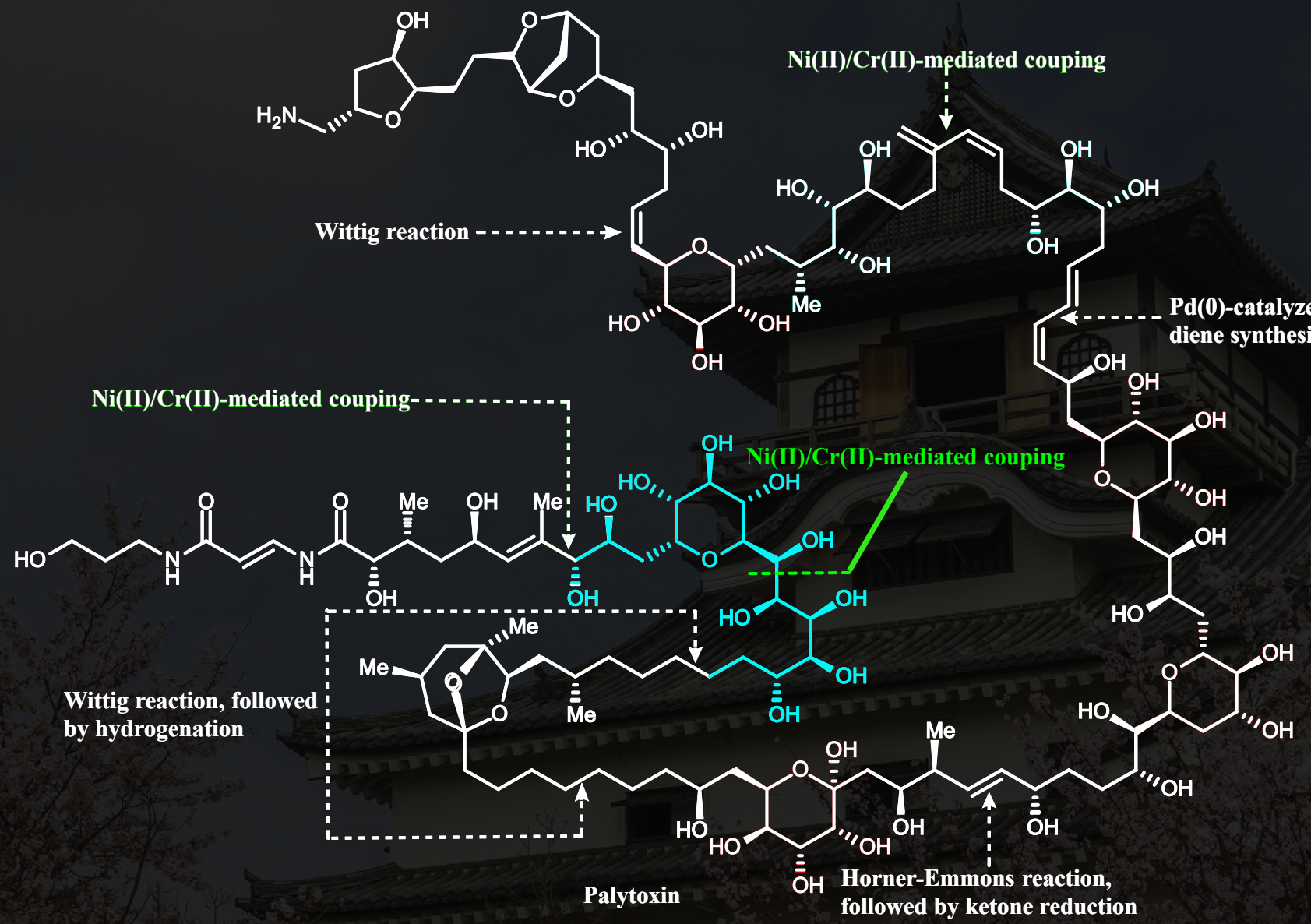
Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

Chemistry from Monensin

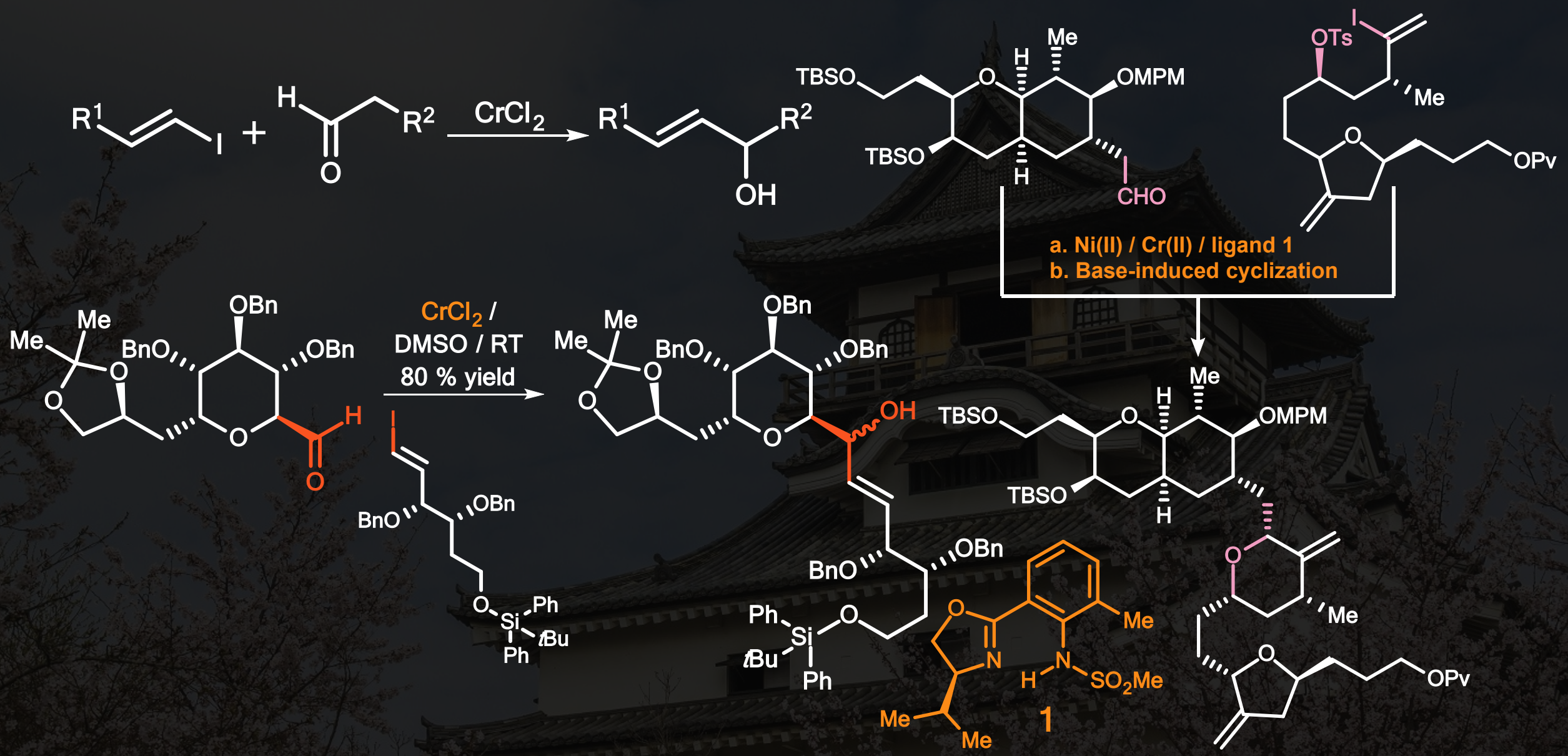


Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

Nozaki - Hiyama - kishi reaction

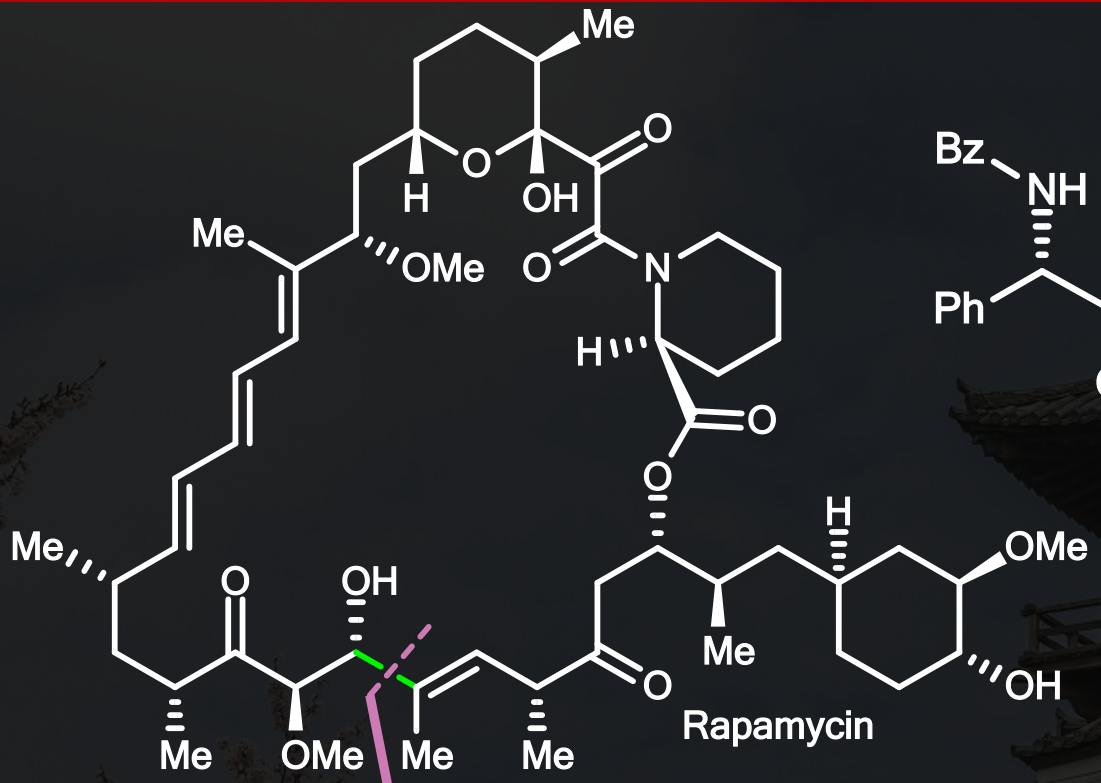


Nozaki - Hiyama - kishi reaction

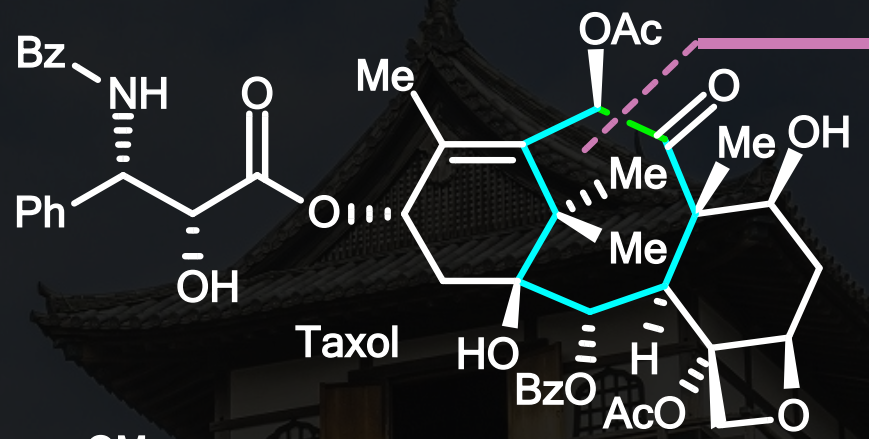


Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

Nozaki - Hiyama - kishi reaction

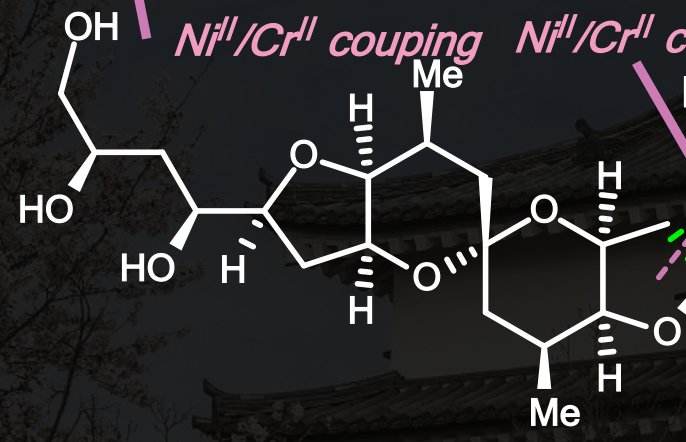


Rapamycin



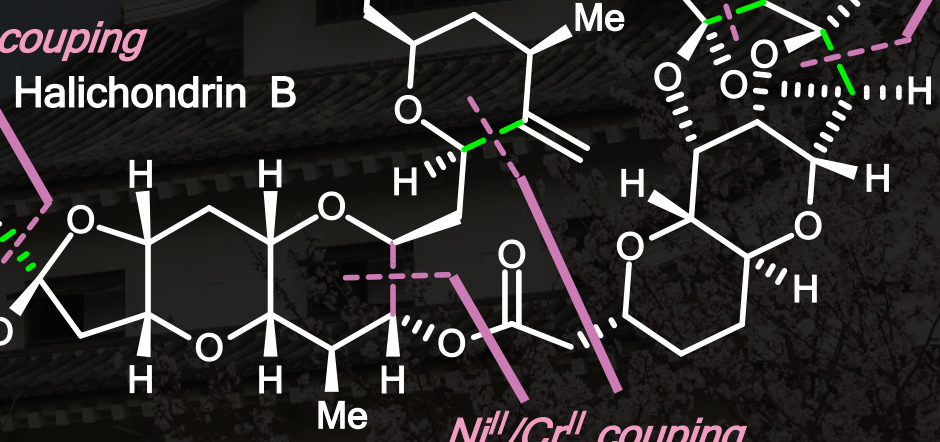
Taxol

Ni^{II}/Cr^{II} couping

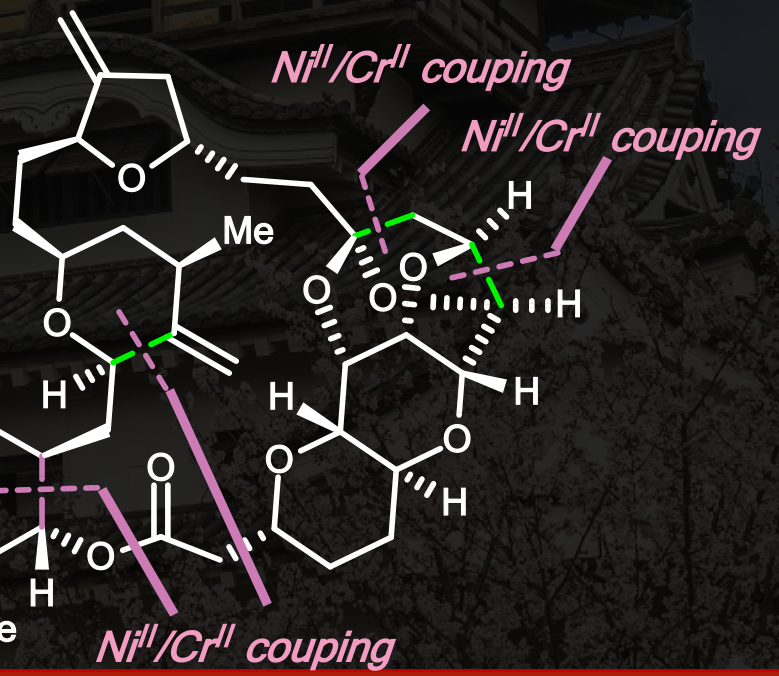


Ni^{II}/Cr^{II} couping

Ni^{II}/Cr^{II} couping



Halichondrin B



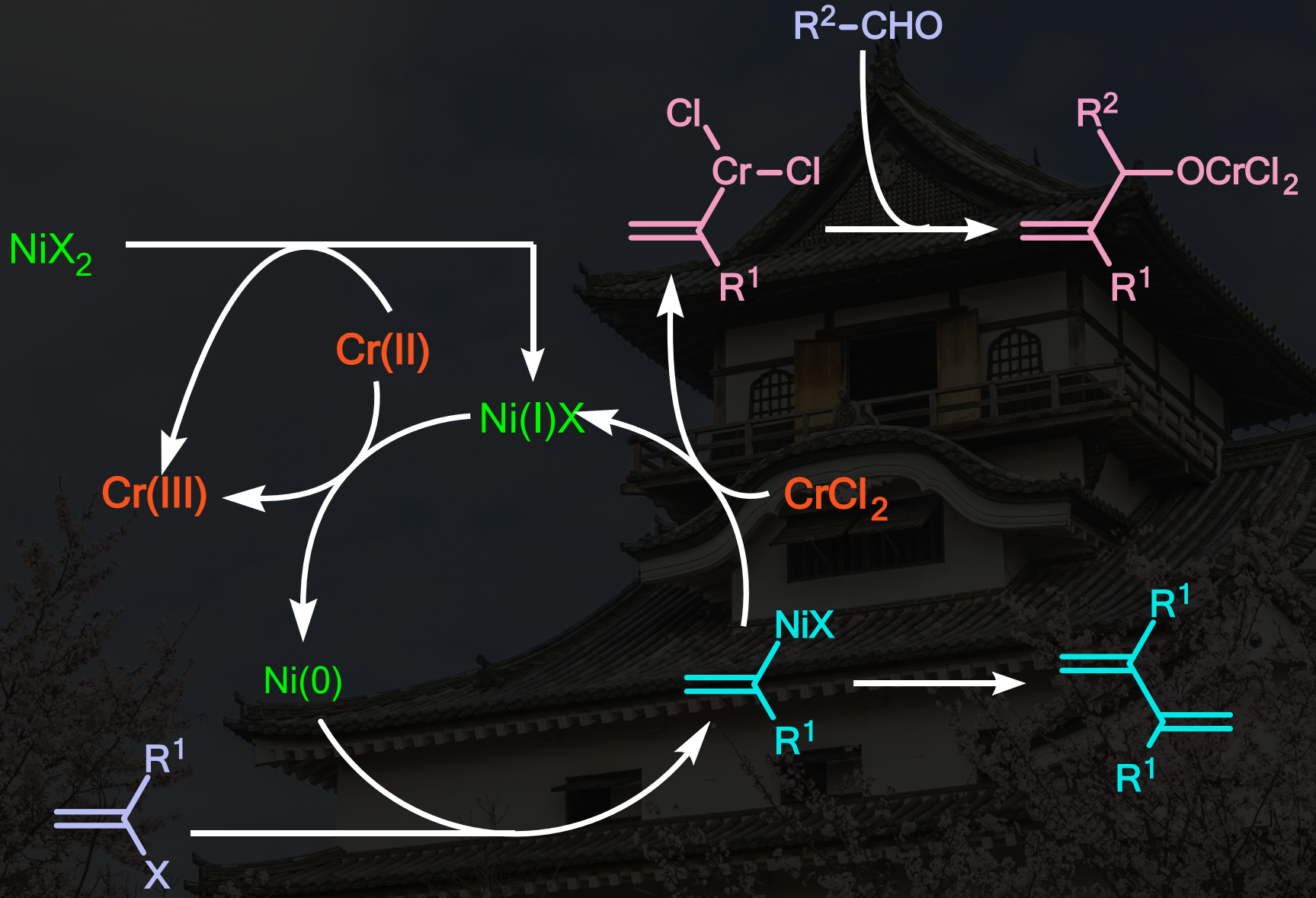
Ni^{II}/Cr^{II} couping

Ni^{II}/Cr^{II} couping

Ni^{II}/Cr^{II} couping

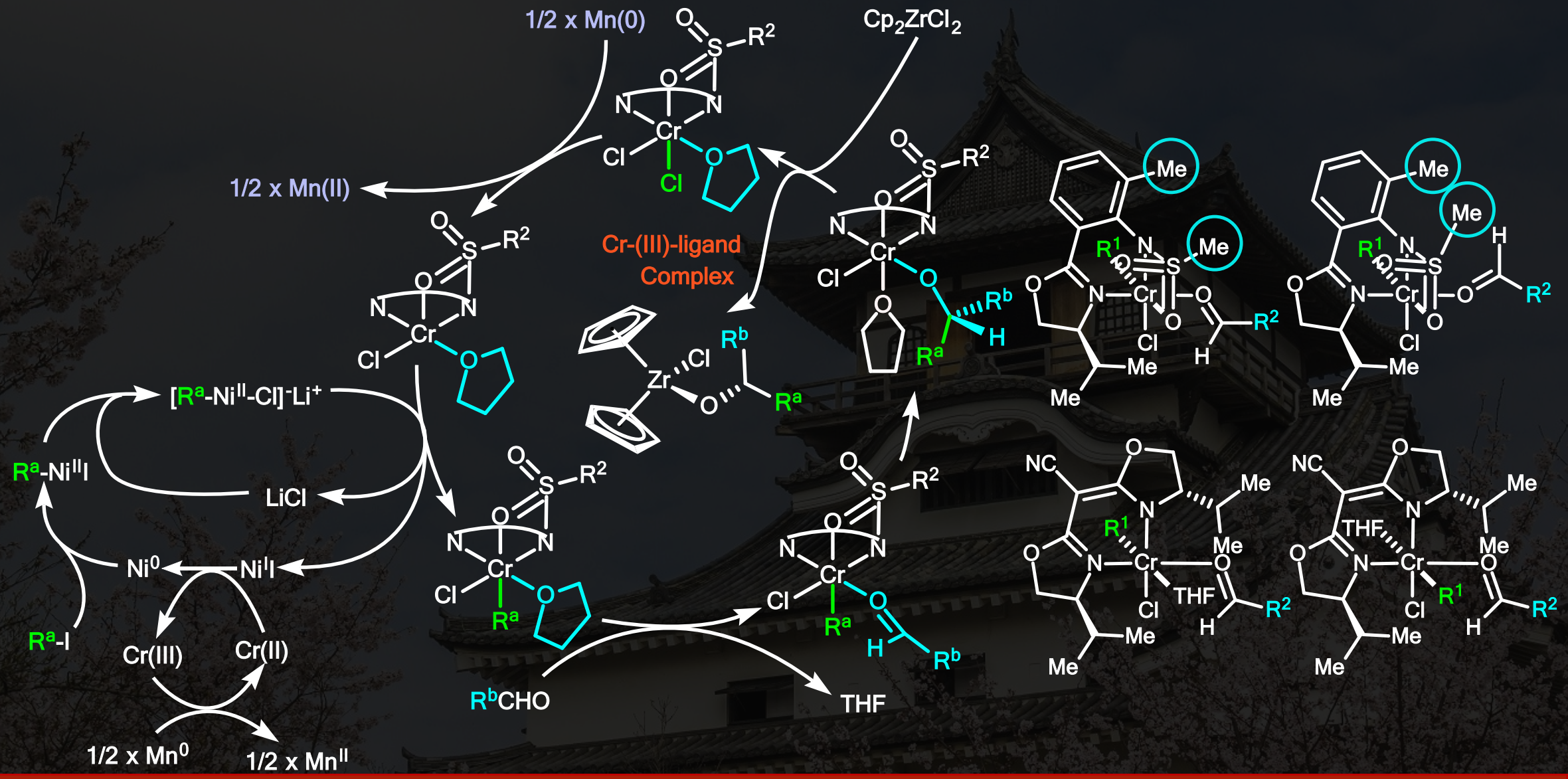
Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

Nozaki - Hiyama - kishi reaction



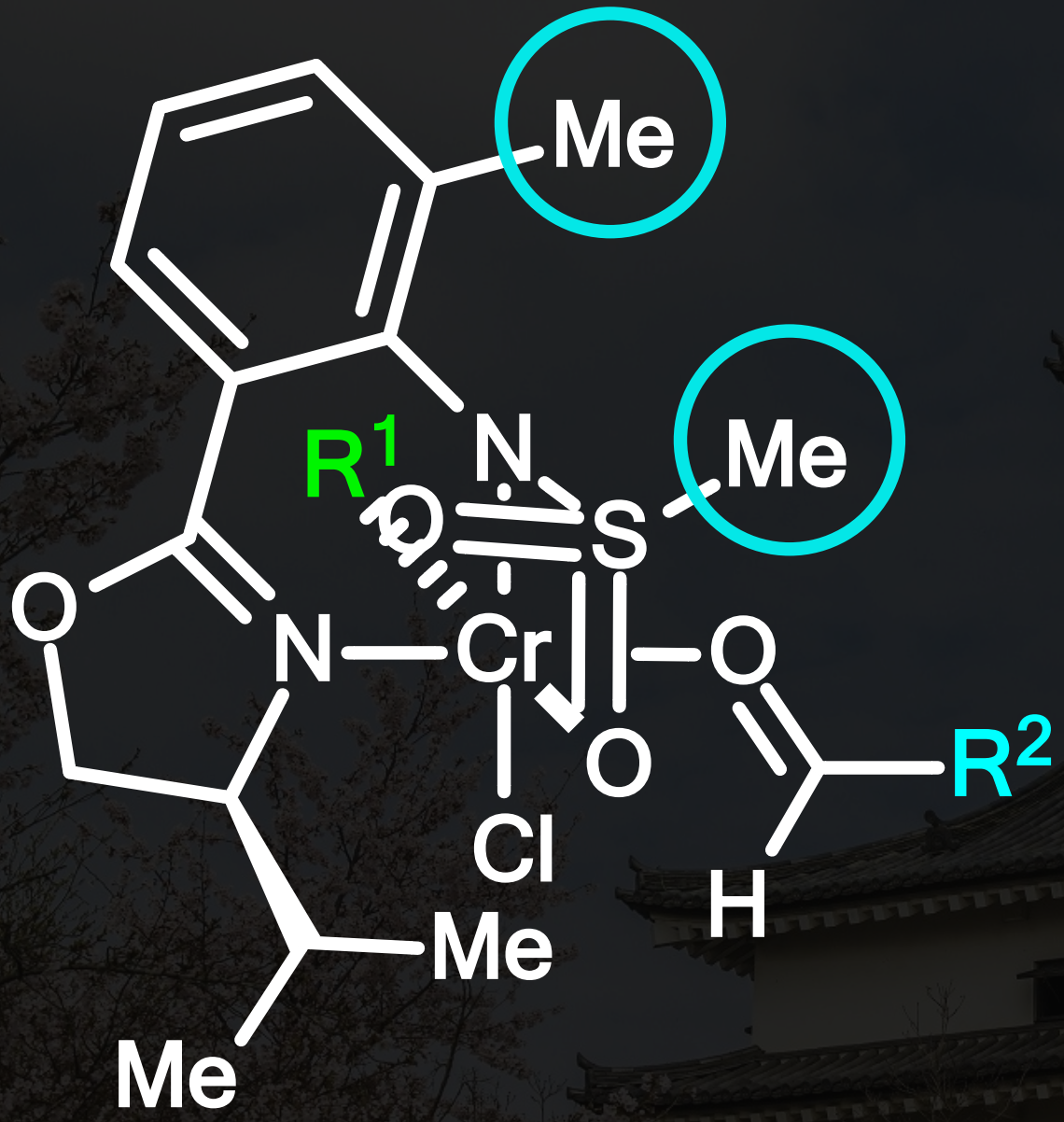
Liu, X.; Henderson, J. A.; Sasaki, T.; Kishi, Y. *Journal of the American Chemical Society* **2009**, *131*, 16678.

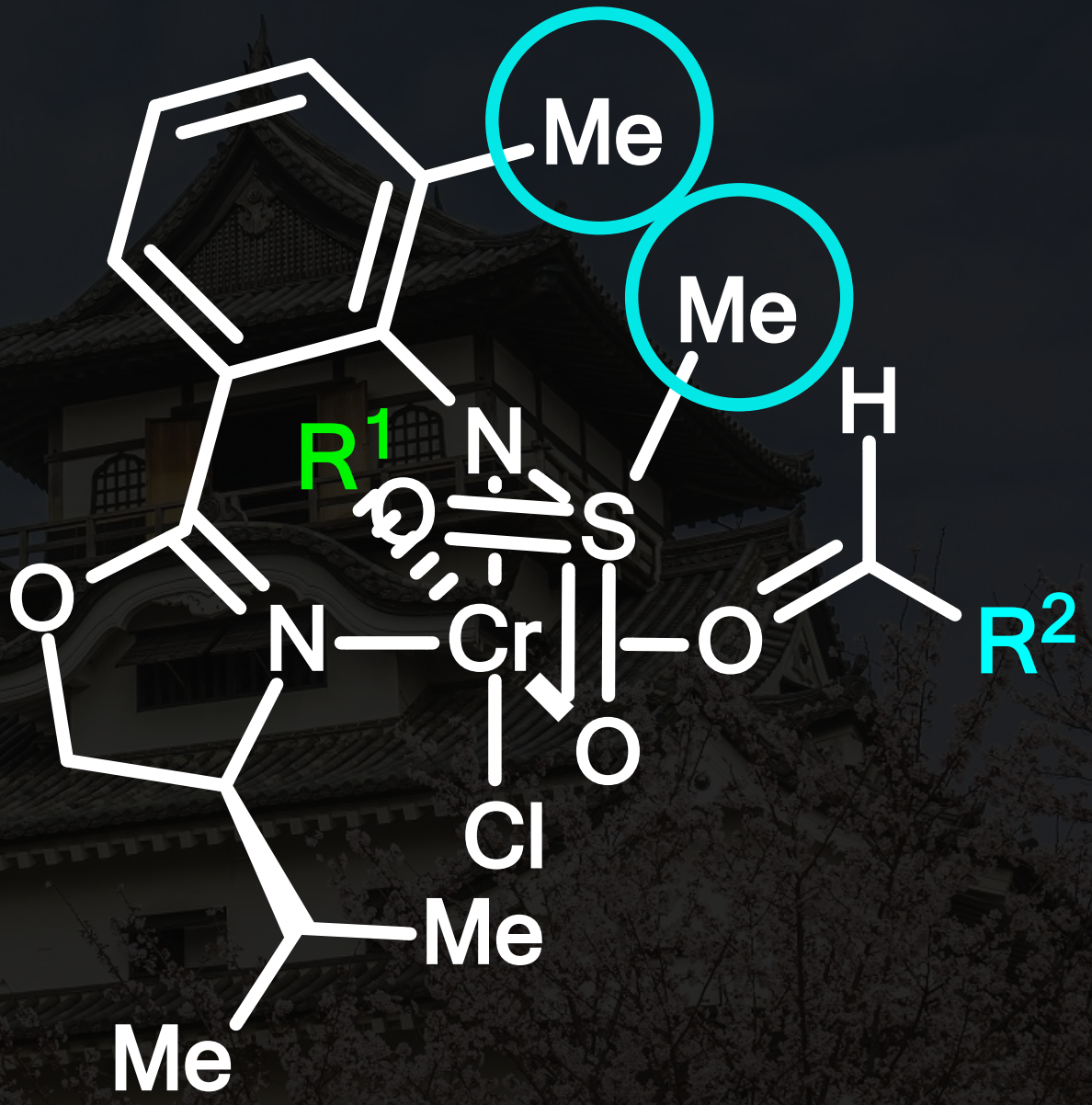
Nozaki - Hiyama - kishi reaction



Guo, H.; Dong, C.-G.; Kim, D.-S.; Urabe, D.; Wang, J.; Kim, J. T.; Liu, X.; Sasaki, T.; Kishi, Y.
Journal of the American Chemical Society **2009**, *131*, 15387.

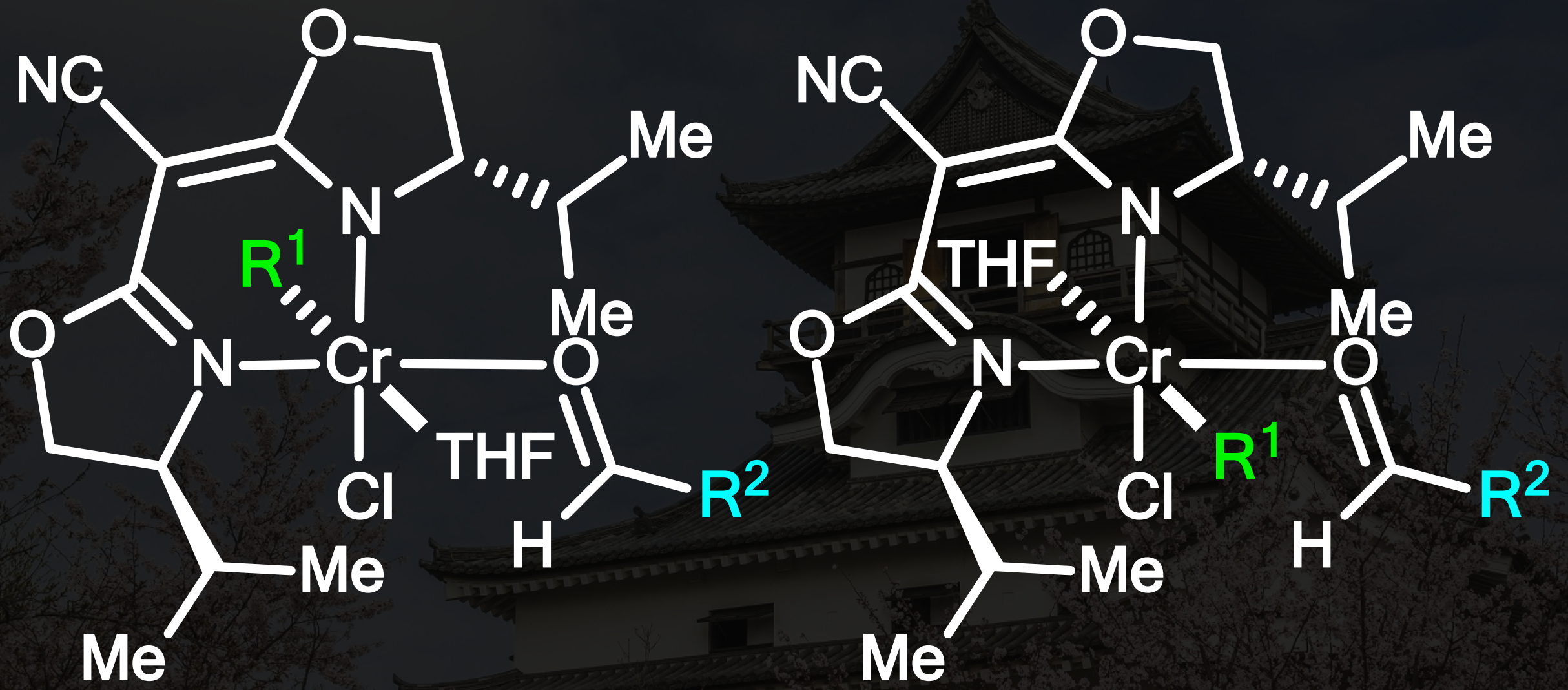
Nozaki - Hiyama - kishi reaction





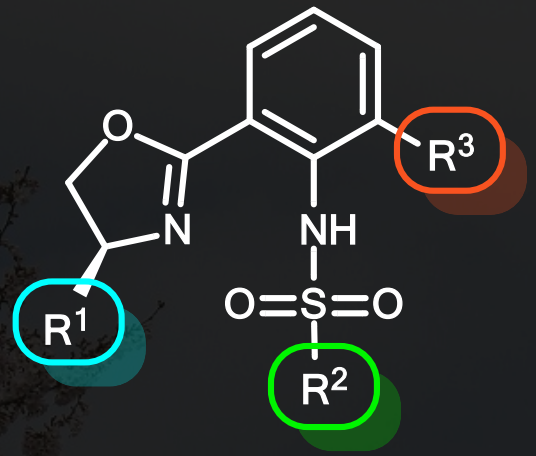
Wan, Z.-K.; Choi, H.-w.; Kang, F.-A.; Nakajima, K.; Demeke, D.; Kishi, Y. *Organic letters* **2002**, *4*, 4431.

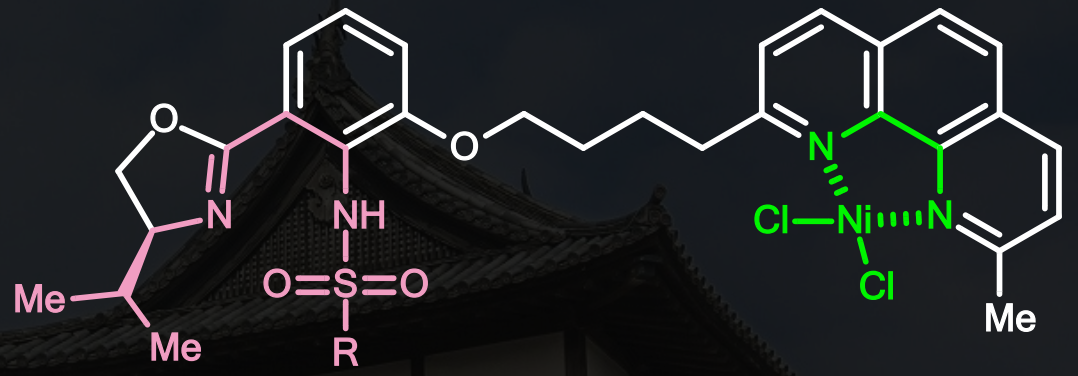
Nozaki - Hiyama - kishi reaction



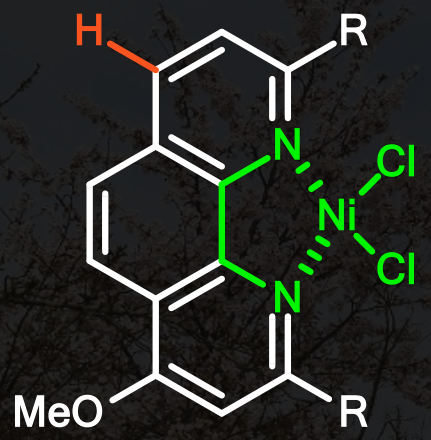
Wan, Z.-K.; Choi, H.-w.; Kang, F.-A.; Nakajima, K.; Demeke, D.; Kishi, Y. *Organic letters* **2002**, *4*, 4431.

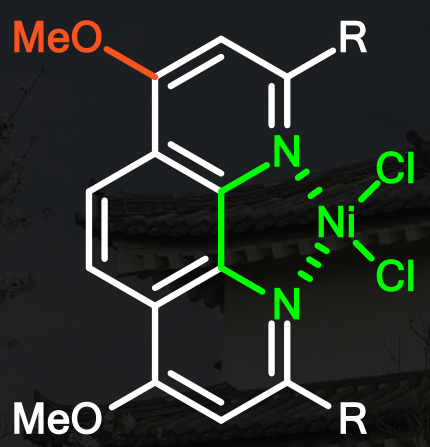
Nozaki - Hiyama - kishi reaction

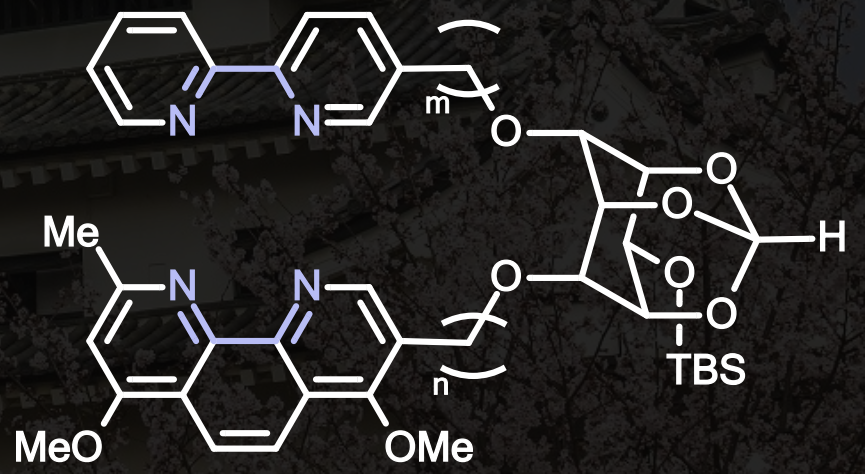




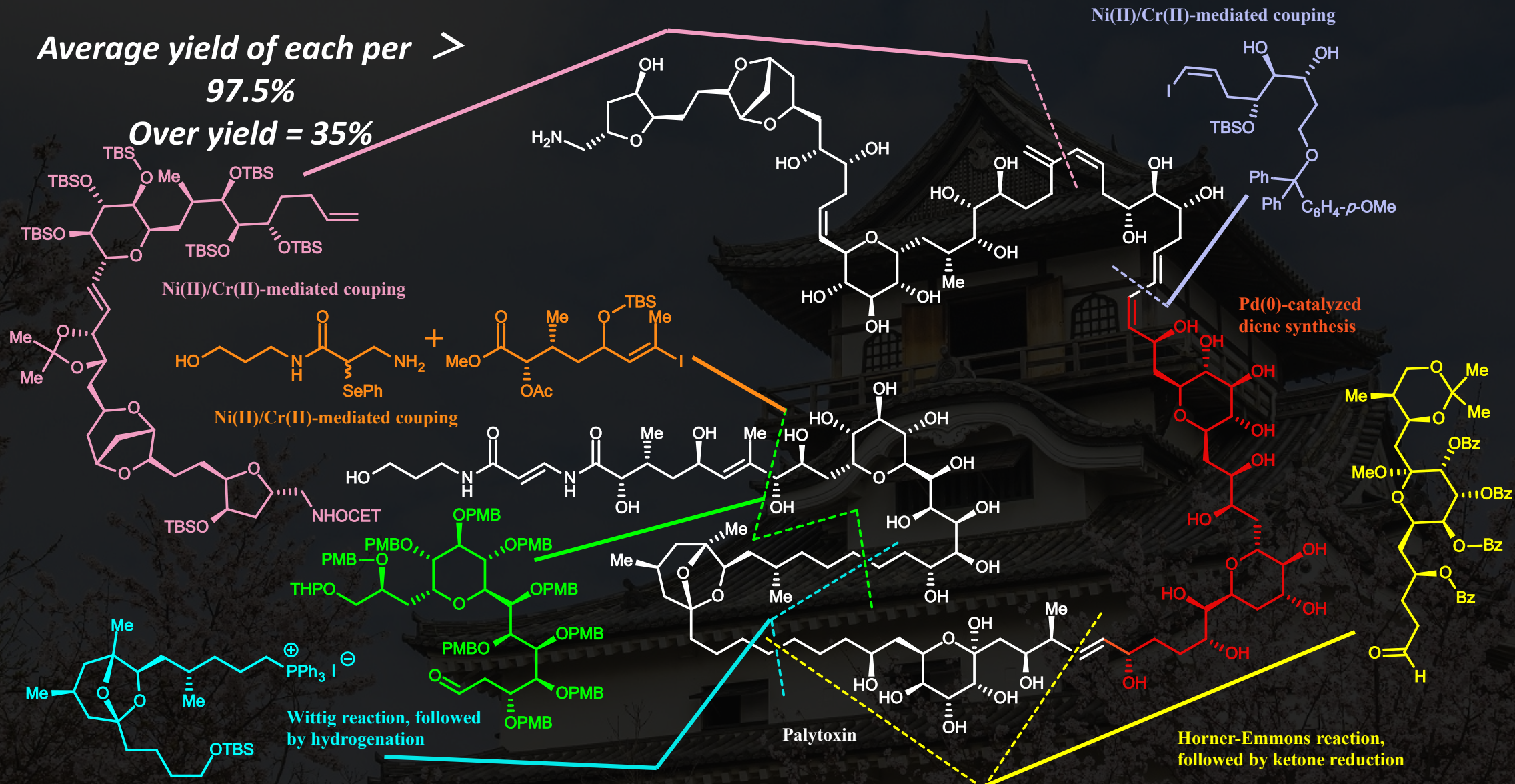






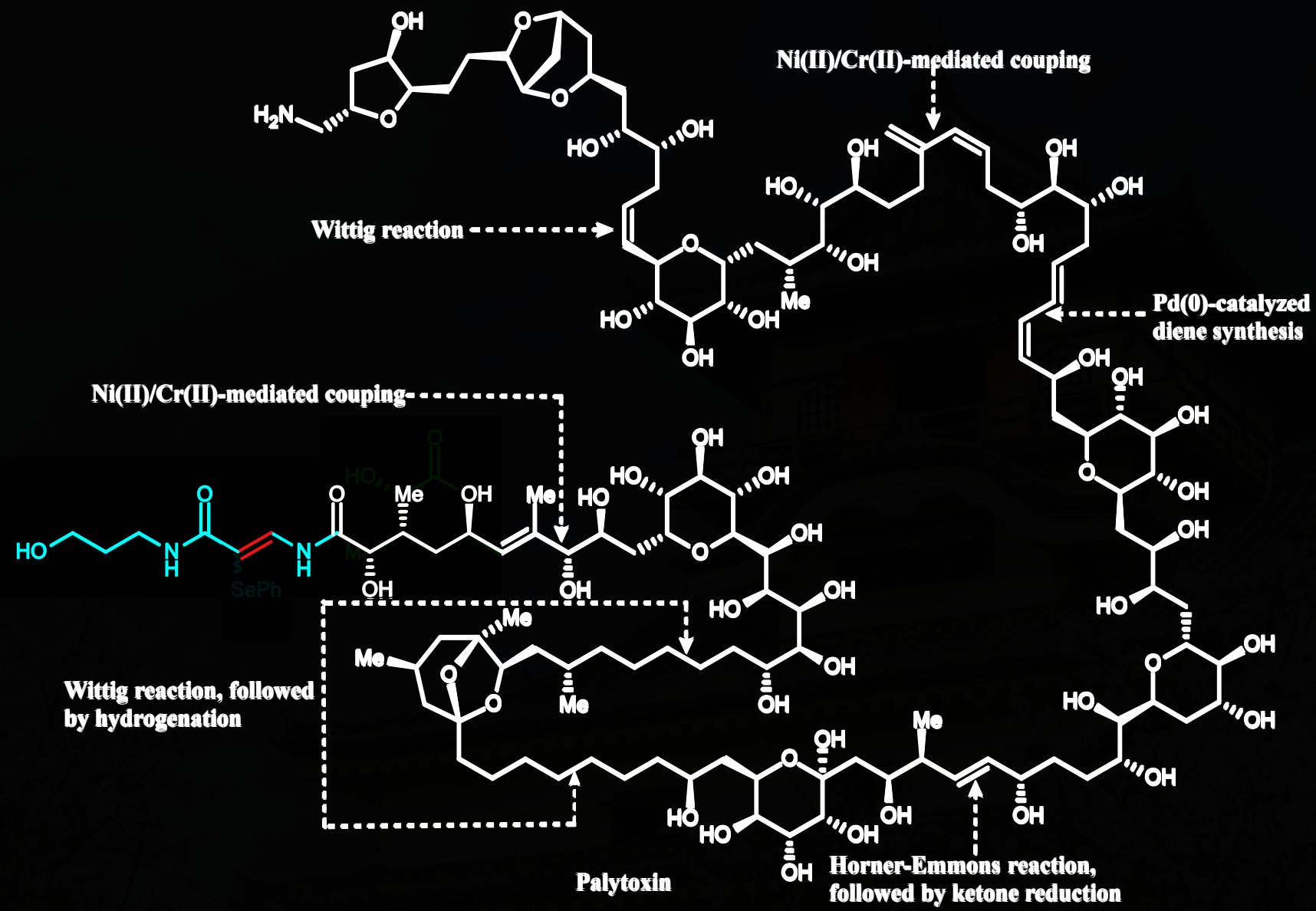


Synthesis of palytoxin



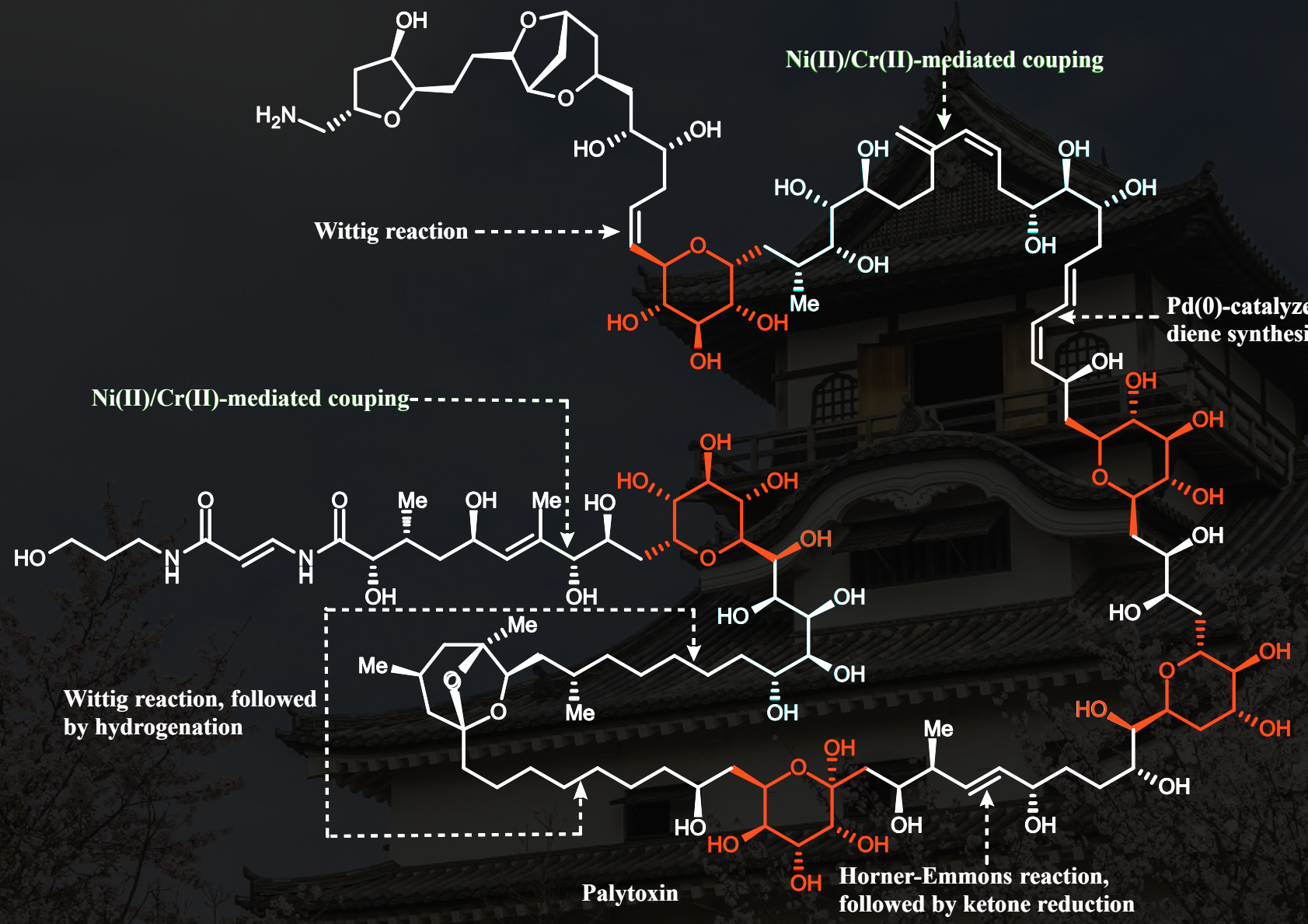
Nicolaou, K. C.; Vourloumis, D.; Winssinger, N.; Baran, P. S. *Angewandte Chemie International Edition* **2000**, *39*, 44.

Synthesis of palytoxin

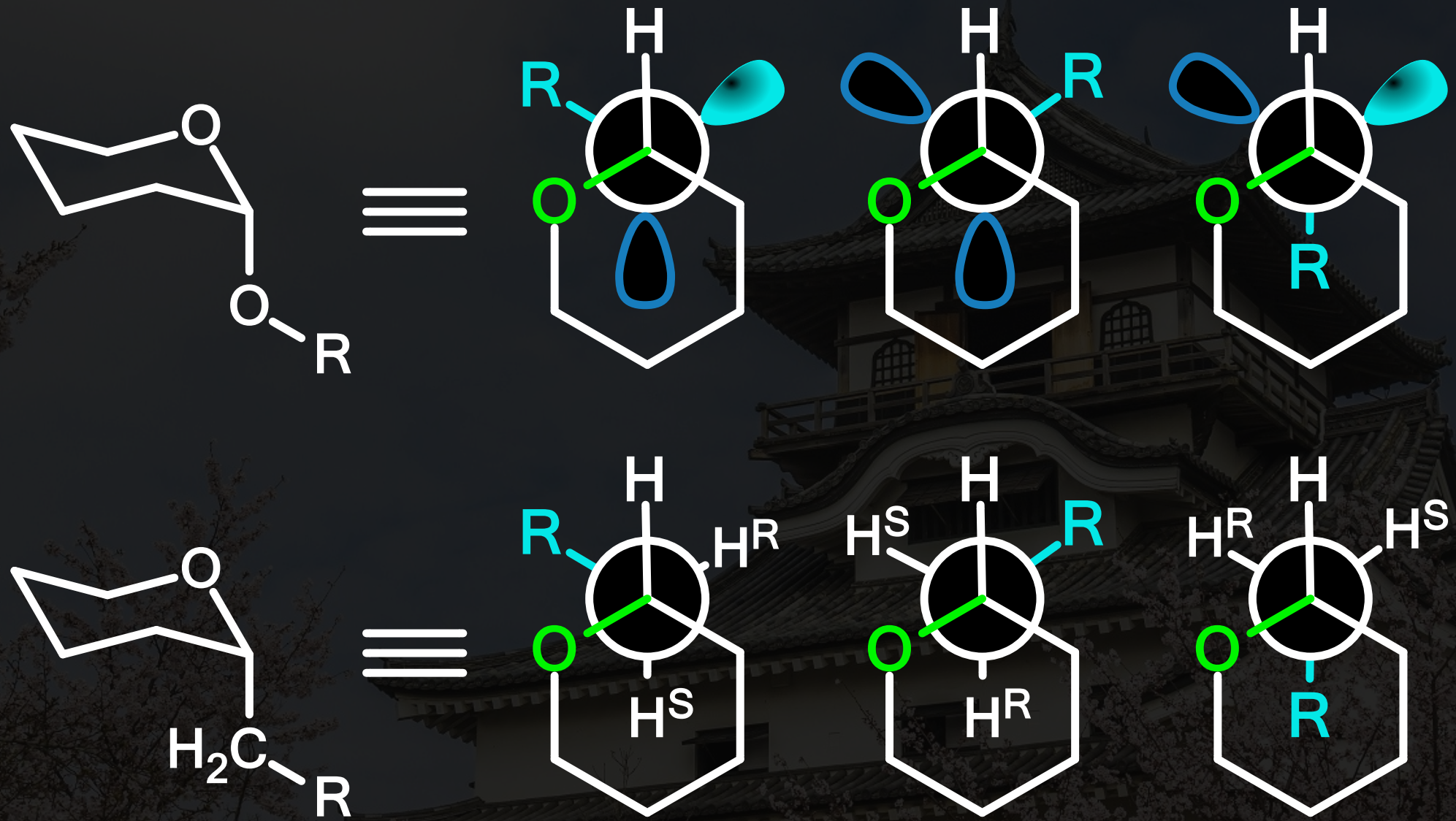


Suh, E. M.; Kishi, Y. *Journal of the American Chemical Society* **1994**, *116*, 11205.

Conformation of glucoside

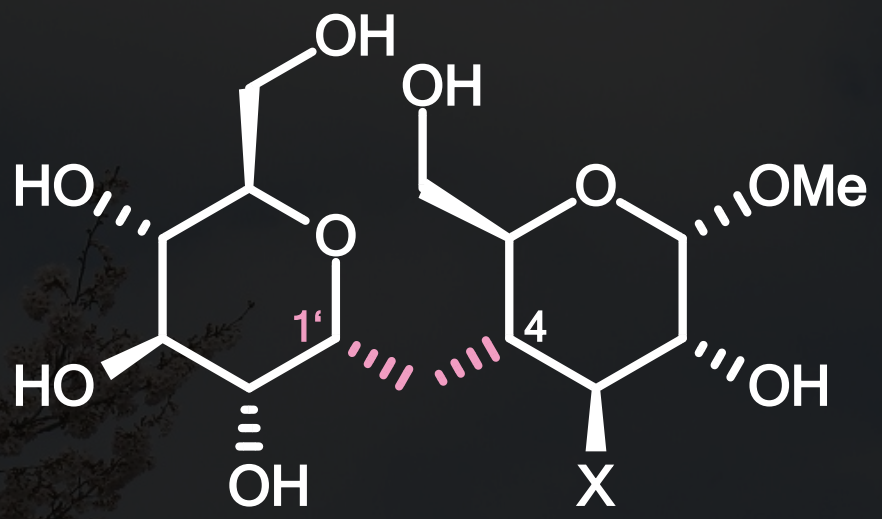


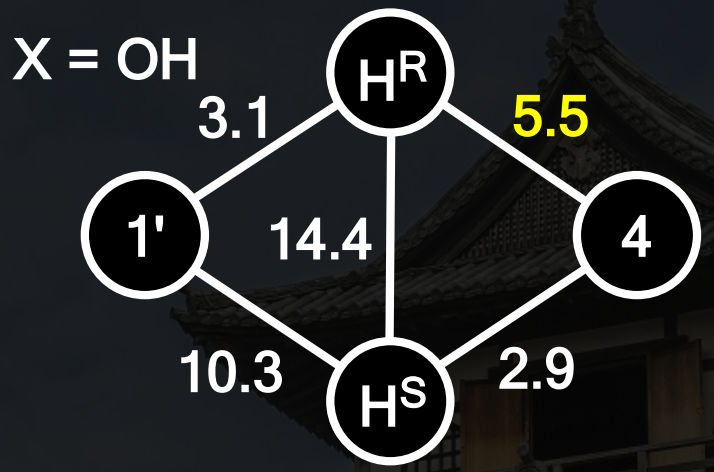
Conformation of glucoside

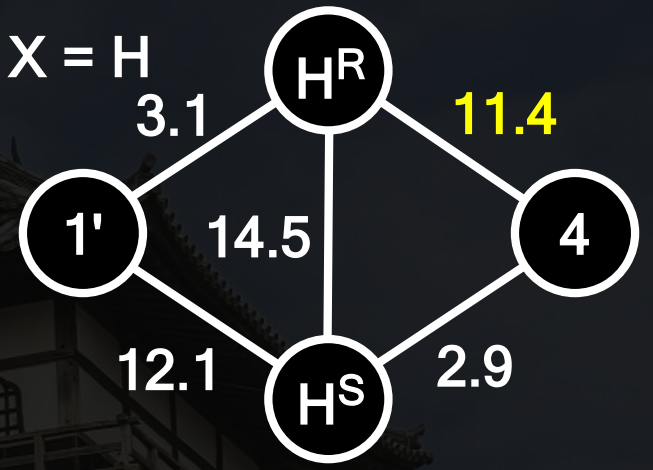


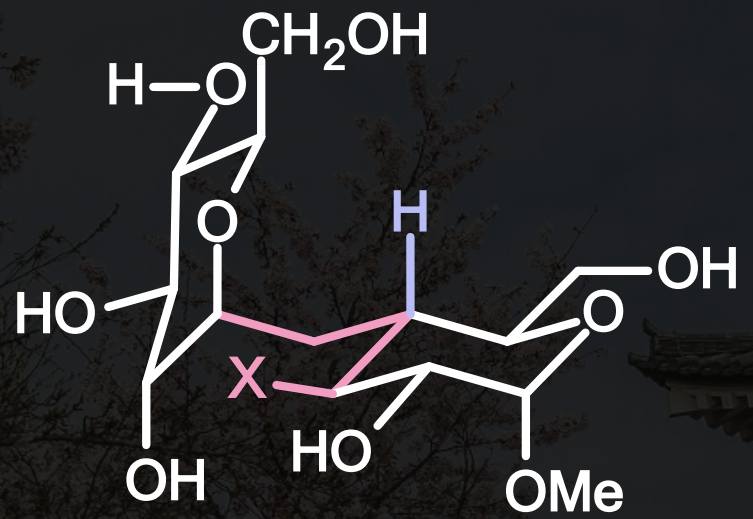
Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

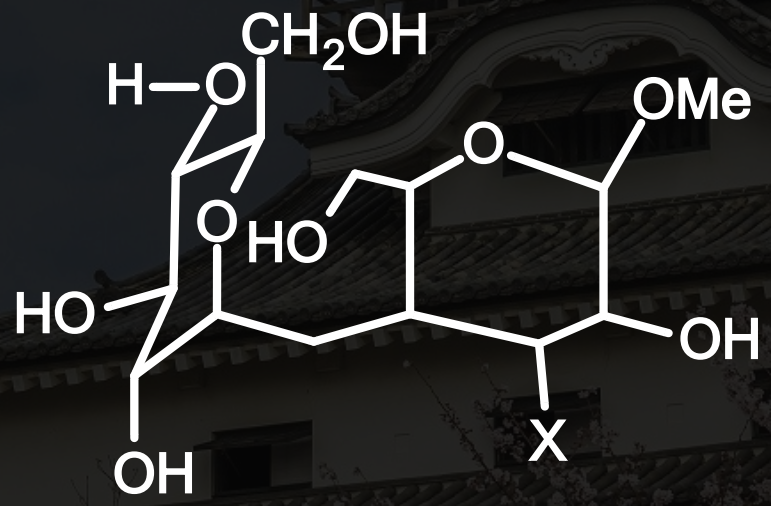
Conformation of glucoside

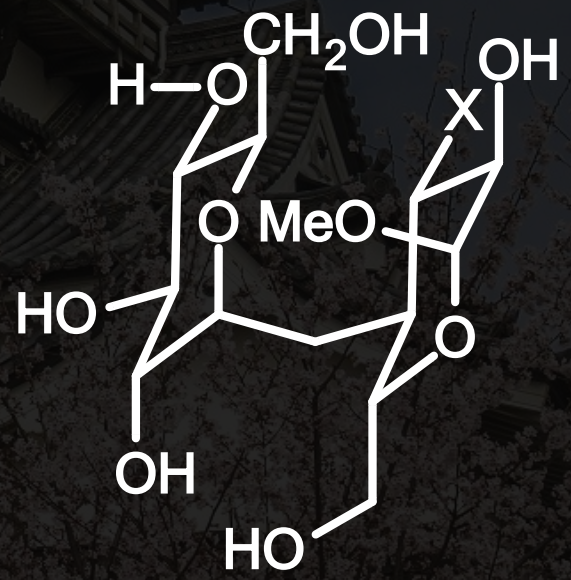






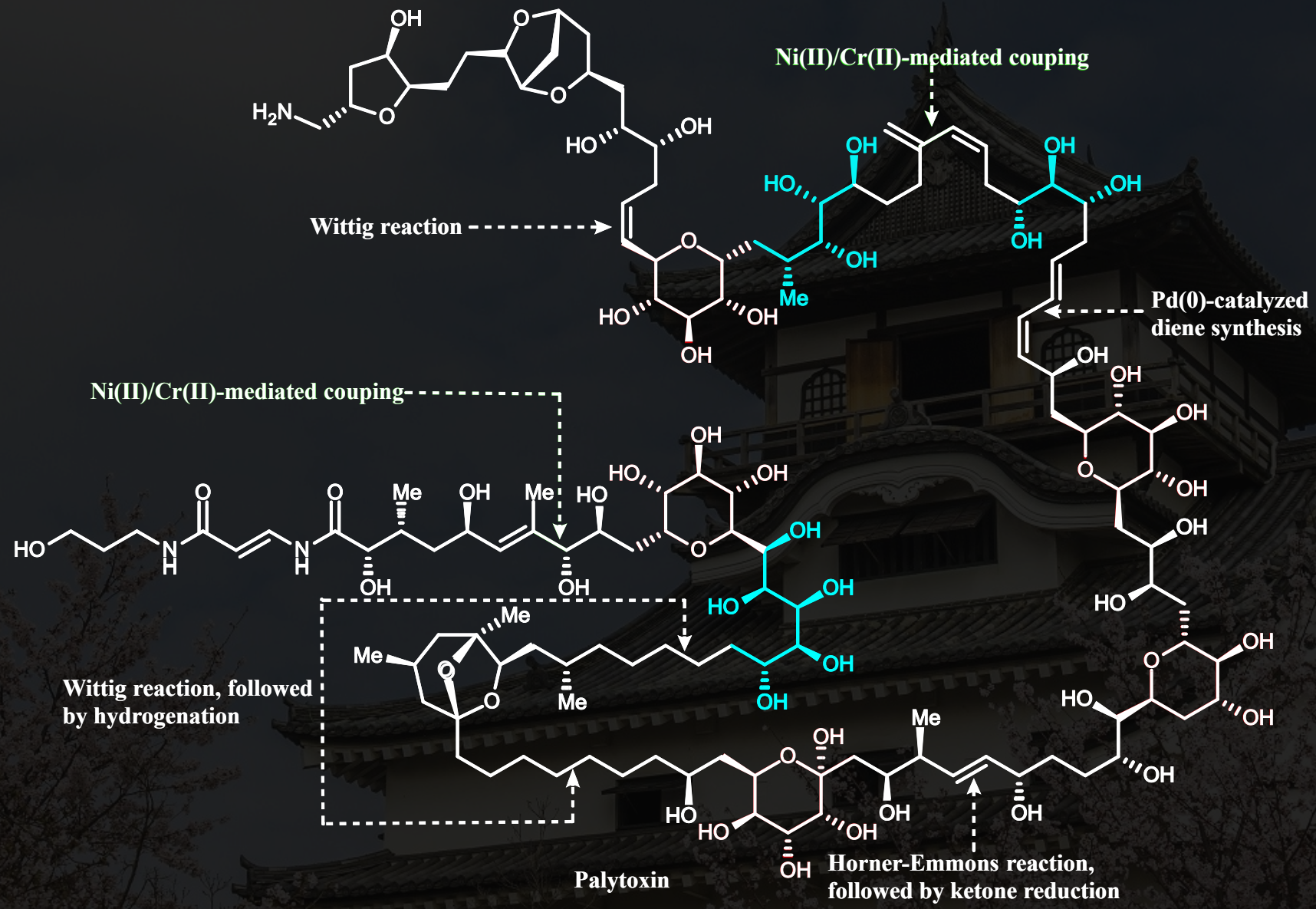




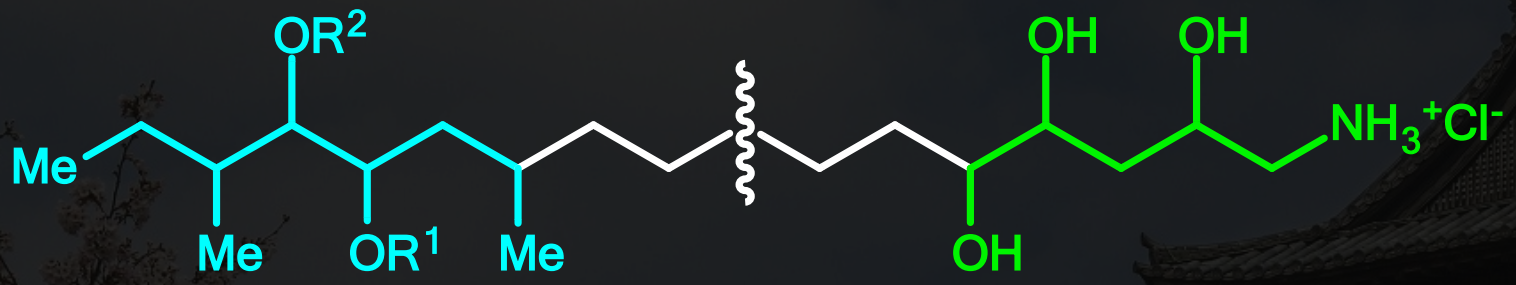


Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

NMR & Configuration



NMR & Configuration



1a : AAL Toxin T_{A1} ; R^1 = tricarballylic acid, R^2 = H

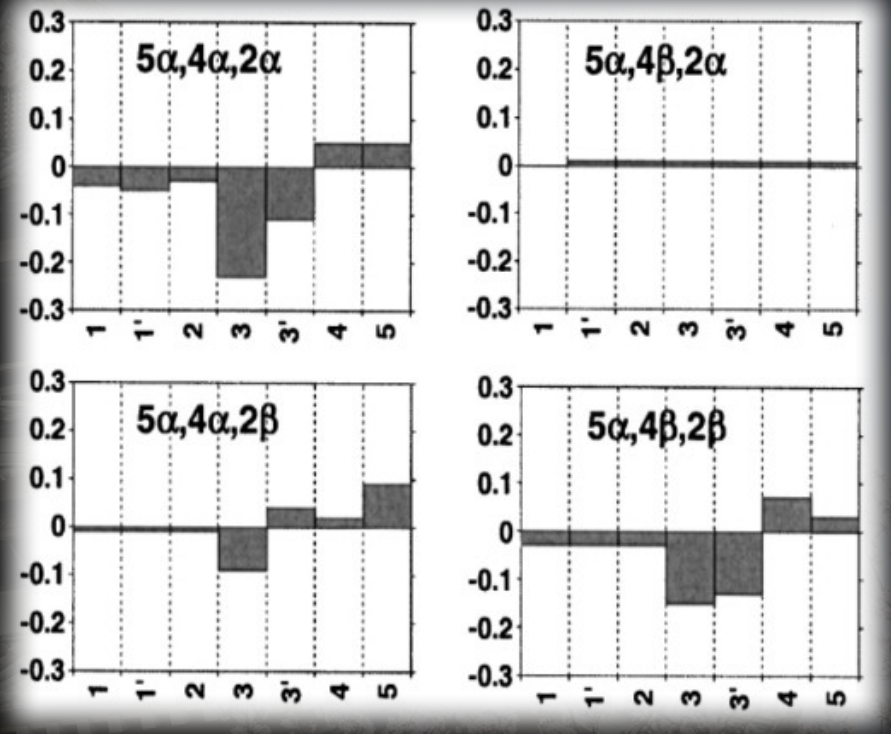
1b : AAL Toxin T_{A2} ; R^1 = H, R^2 = tricarballylic acid

2 : AAL Toxin T_A backbone : $R^1 = R^2$ = H



Right-Half Model of the backbone of AAL Toxin T_A

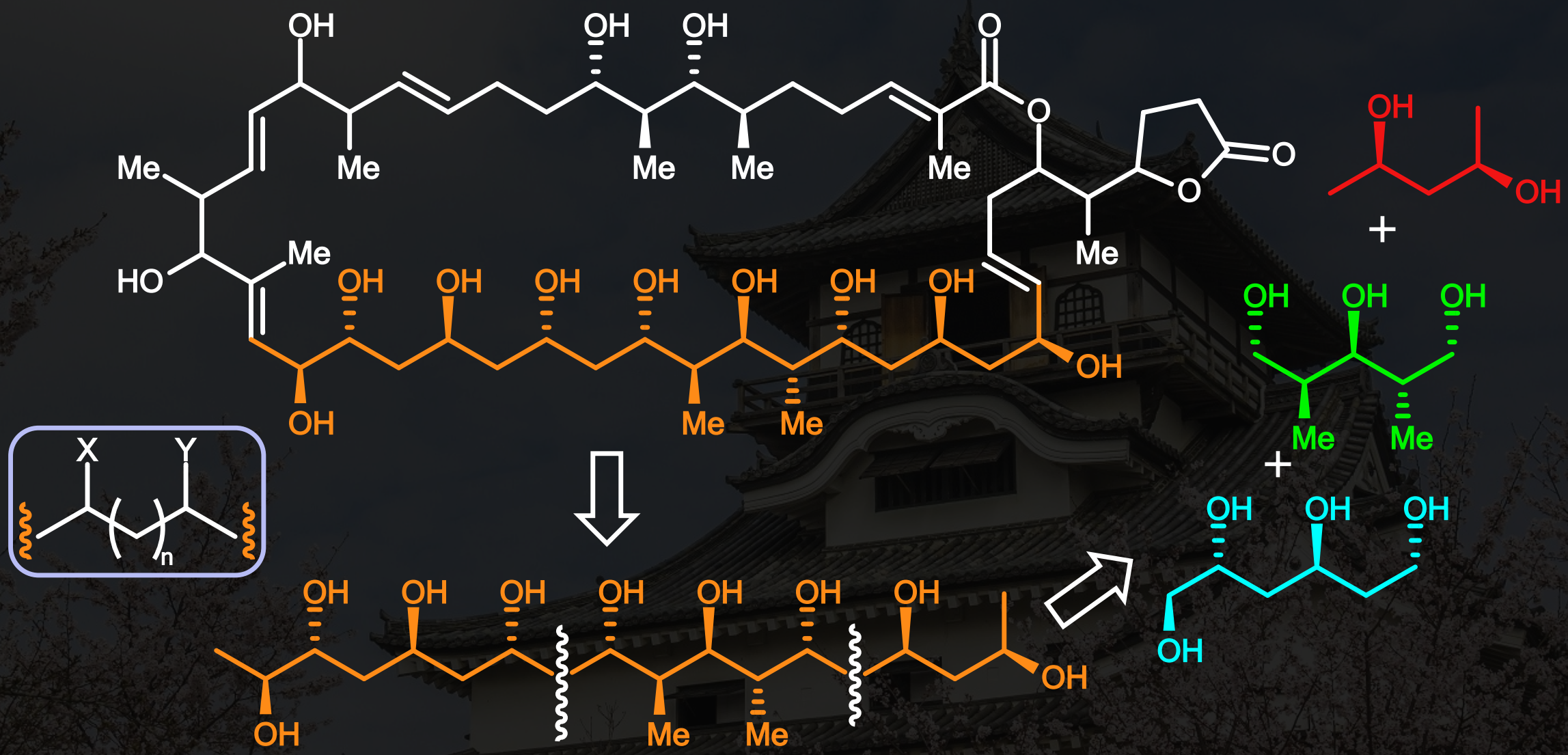
5α,4β,2α–diastereomer



Comparison of ^1H chemical shifts ($\delta_{AAL\ toxin} - \delta_{SYN}$ in ppm).

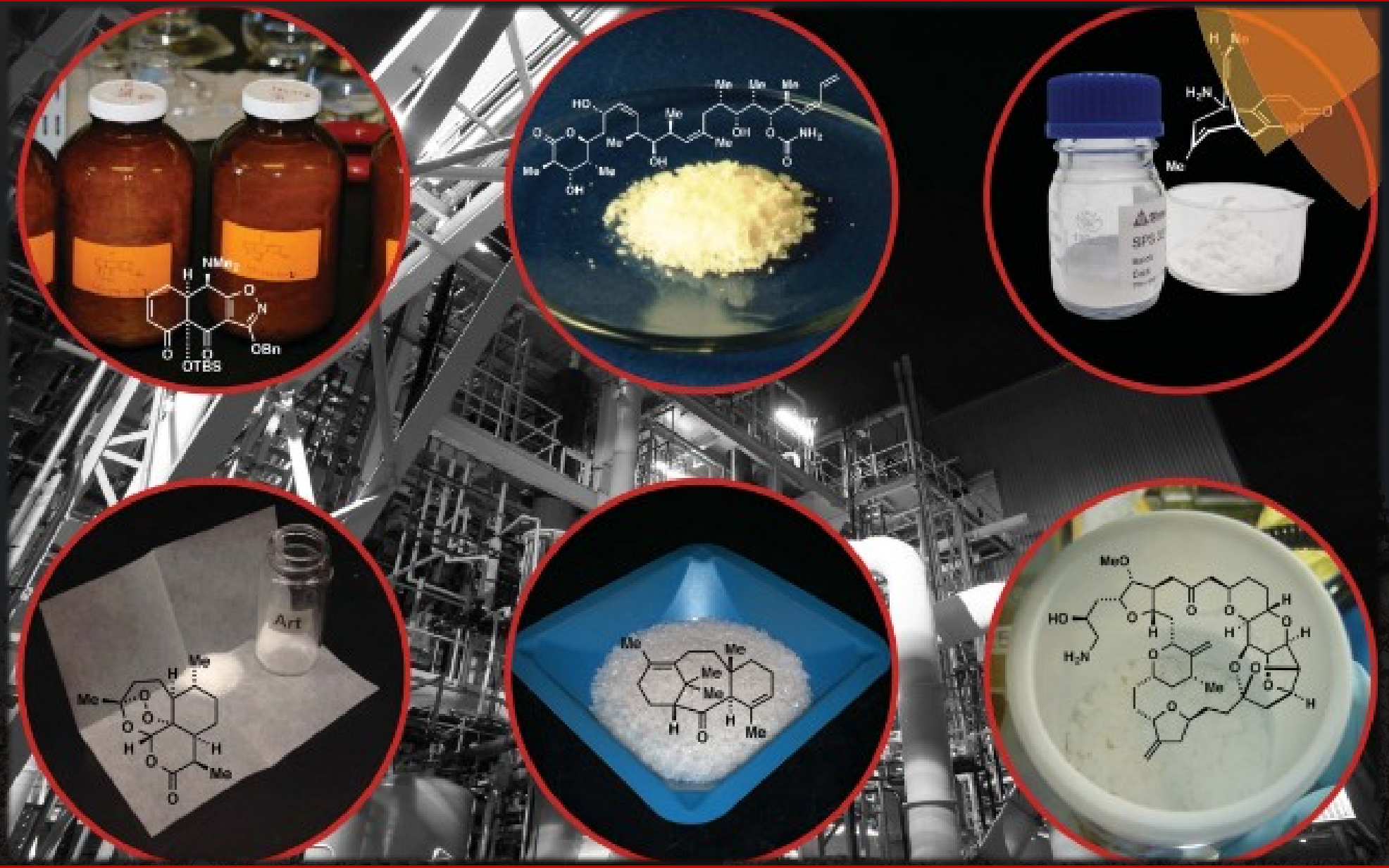
Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

NMR & Configuration



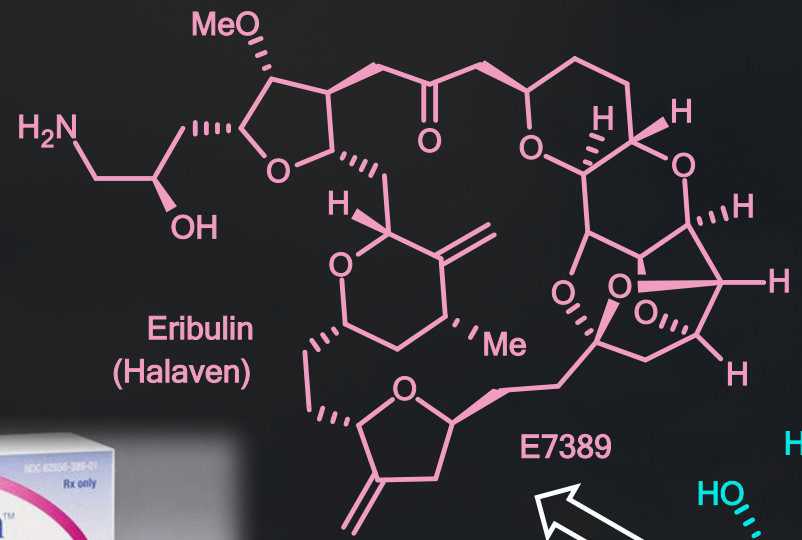
Kishi, Y. *Tetrahedron* **2002**, *58*, 6239.

Medicine & Halichondrin B



Kuttruff, C. A.; Eastgate, M. D.; Baran, P. S. *Natural Product Reports* **2014**, *31*, 419.

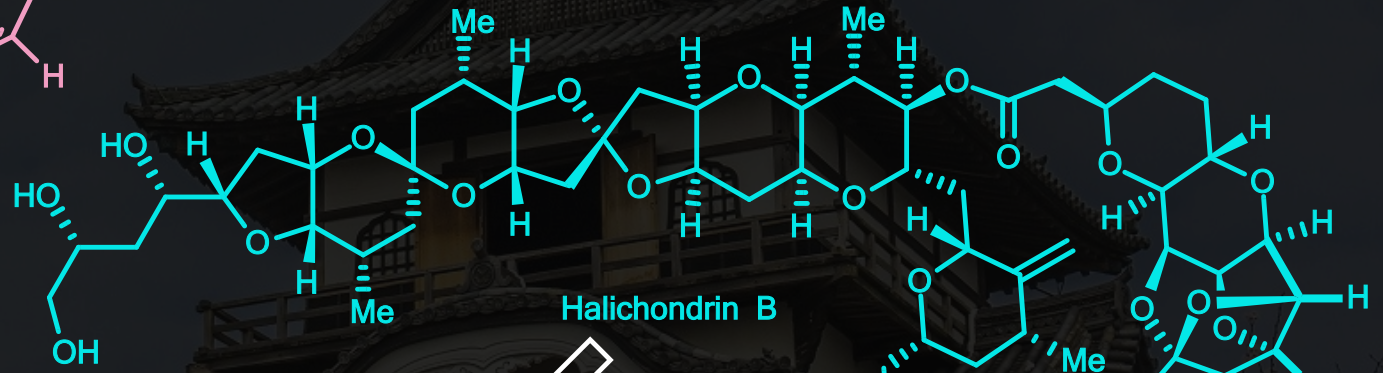
Medicine & Halichondrin B



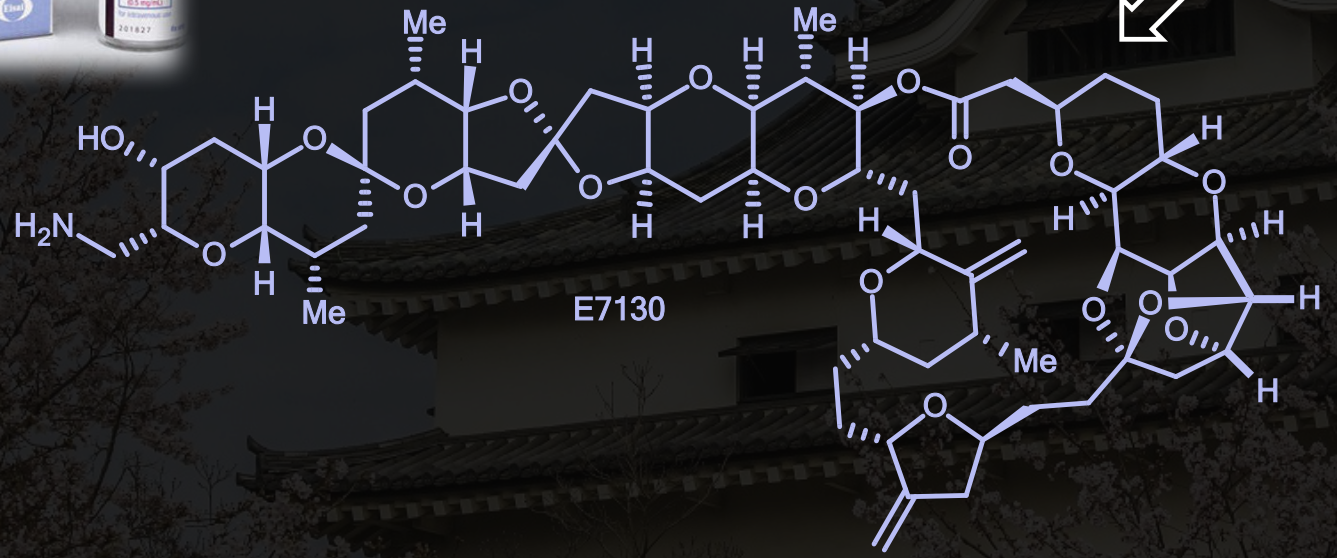
Eribulin
(Halaven)



E7389

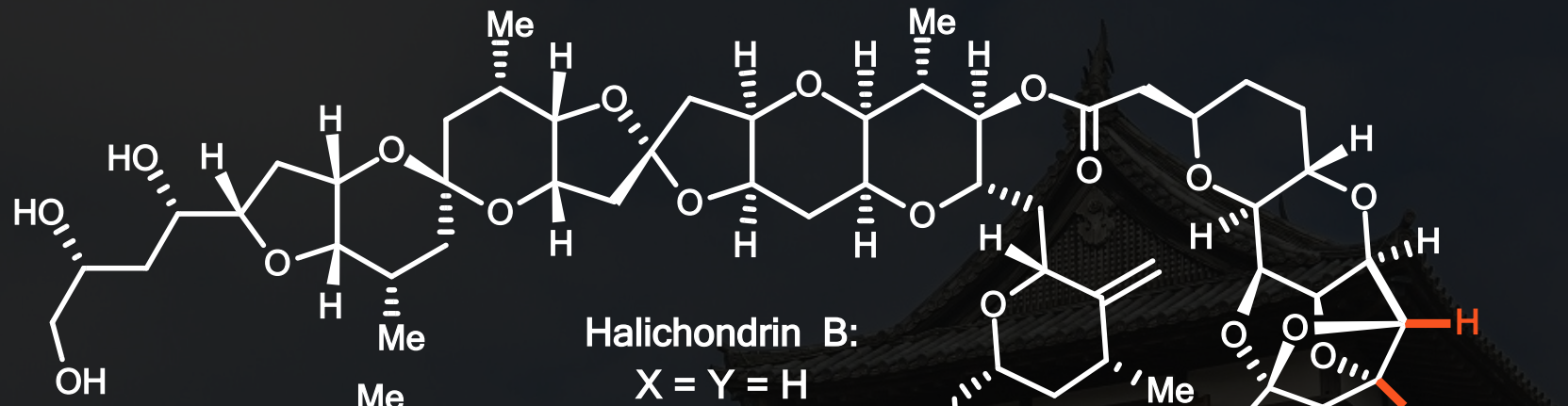


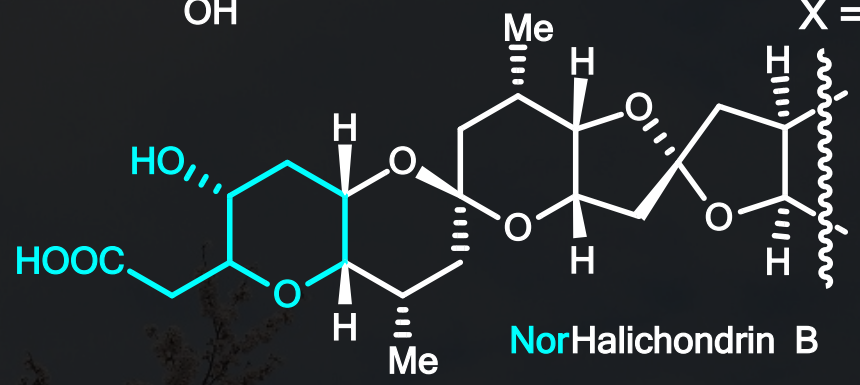
Halichondrin B

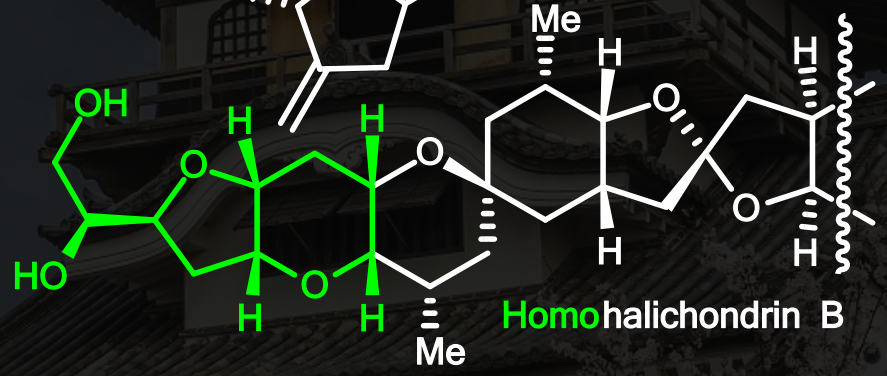


E7130

Recent developments



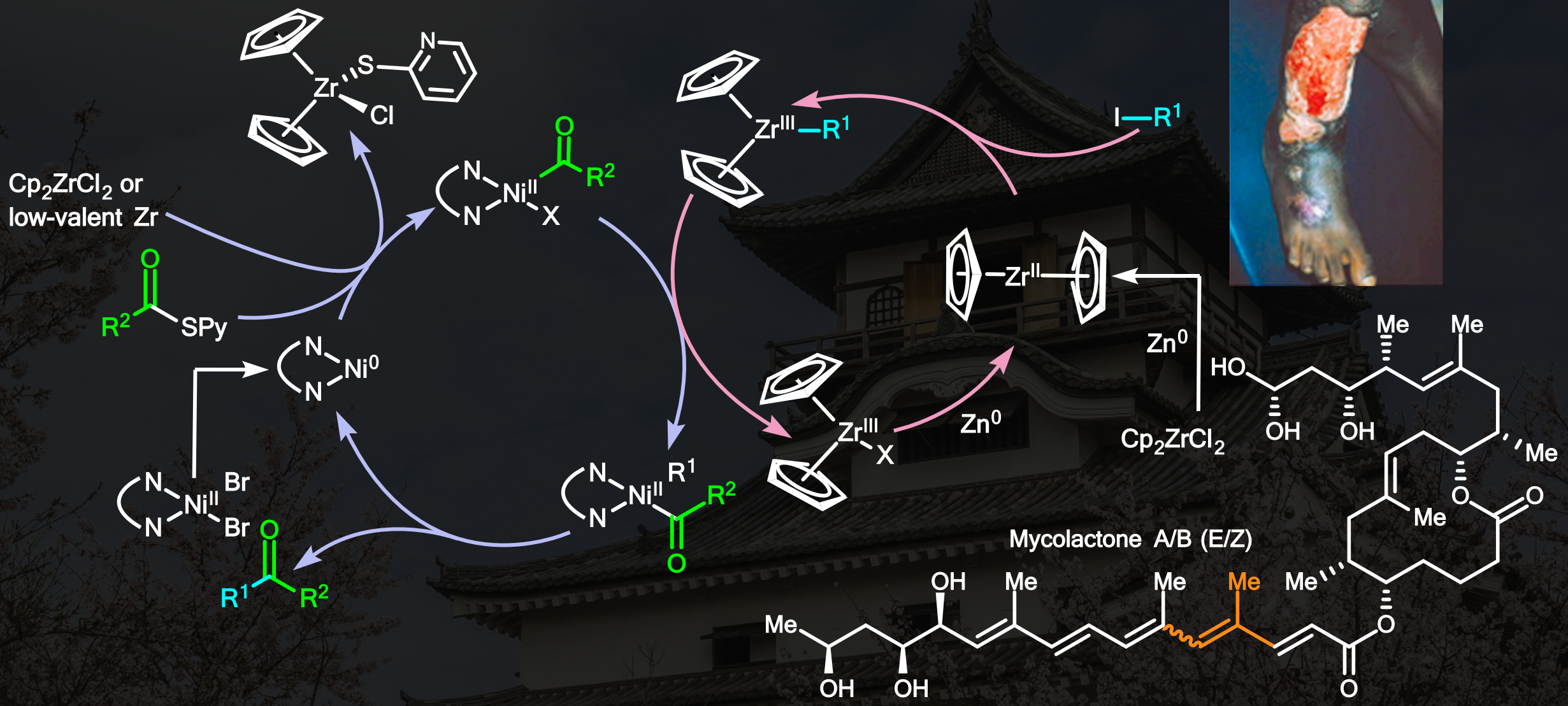




	A series (X = Y = OH)	B series (X = Y = H)	C series (X = OH; Y = H)
Halichondrin	Note	known	known
Norhalichondrin	known	known	known
Homohalichondrin	konwn	known	known

Yahata, K.; Ye, N.; Iso, K.; Naini, S. R.; Yamashita, S.; Ai, Y.; Kishi, Y. *The Journal of organic chemistry* **2017**, *82*, 8792.

Recent developments



Ai, Y.; Ye, N.; Wang, Q.; Yahata, K.; Kishi, Y. *Angewandte Chemie International Edition* **2017**, *56*, 10791.

Kishi, Y. *Proceedings of the National Academy of Sciences* **2011**, *108*, 6703.